

Response to Reviewers

Dear Editor,

We sincerely thank the editor and reviewers for their careful reading of our manuscript and for the constructive comments. We have revised the manuscript, Supplementary Information, figures, tables, equations, references, and code/data availability statement to make the data-generation workflow, optimizer formulation, comparison protocol, physical stress-test boundary, and climate-sensitivity boundary explicit.

This response letter has been reorganized as a comment-level evidence package. Each reviewer comment is reproduced under its own comment heading. The response then states the revision, identifies where the manuscript was changed, and displays the relevant revised manuscript, Supplementary Information, figure, table, equation, or local reference material directly under that comment.

Manuscript title: Surrogate-conditioned optimizer ranking in block-scale urban energy design: A comparison of policy learning and Pareto search

Manuscript ID: APEN-D-26-08148

Sincerely,

The Authors

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Summary of major revisions

We are grateful to the editor and reviewers for the careful and constructive evaluation of our manuscript. After rereading the original submission in light of the comments, we recognize that several parts of the previous version were not sufficiently clear, especially the data-generation workflow, the role of the analytic response surface, the DDPG formulation, the optimizer comparison protocol, and the interpretation of the physics-based and climate-sensitive calculations. We apologize for the additional effort that these ambiguities caused during review. The comments were highly valuable and led us to revise the manuscript, figures, tables, equations, Supplementary Information, and reproducibility statement in a much more transparent form.

The revised manuscript is therefore not a minor language edit. We substantially rewrote the study framing, clarified the distinction between generated feasible morphologies and morphology descriptors, replaced ambiguous validation language with a limited physics-based cross-model stress test, added cross-climate sensitivity calculations, corrected the reward formulation, and reorganized the Supplementary Information so that the equations, validation protocols, optimizer analyses, physical case results, and climate case results are directly traceable.

Table R1: Editor-facing overview of the main revision packages.

Revision package	Comments addressed	Manuscript revision	Supporting material
Framing and readability	R1-1, R1-2, R1-15, R4-1	Abstract, Introduction, Nomenclature, Discussion, Conclusions	SI S1–S6 aligned with revised terminology
Data source and objective definitions	R2-1, R2-6, R2-7, R3-5	Analytic response generation, evaluation modes, target definitions	SI S1 equations, clipping ranges, descriptor dependencies
Sampling and surrogate assessment	R1-7, R1-8, R2-4	Sampling protocol, descriptor coverage, repeated validation	SI S1–S2, Table S2, Fig. S1–S3
DDPG and optimizer comparison	R1-10, R1-11, R1-13, R2-5, R3-2, R3-3, R4-4	Static surrogate-query DDPG, reward correction, output contracts	SI S3, equal-size metrics, HV ceiling analysis
Feasible projection and physical stress test	R1-7, R2-2, R2-3, R3-5	Nearest-feasible-block projection and 24-case cross-model stress test	SI S5, projection metrics, physical summary table
Cross-climate and reproducibility	R1-14, R2-8, R4-2, R4-3, R4-5	Four-block climate sensitivity, future energy-system scope, data/code statement	SI S6, weather metadata, public repository statement

Table R1 is only an editor-facing overview. The evidence for each reviewer comment appears under the corresponding comment below, where revised manuscript excerpts, SI excerpts, figures, compact tables, equations, and local references are shown next to the response that uses them.

► **Reviewer 1.**

We thank Reviewer 1 for the detailed comments on readability, terminology, figures, software context, parameter selection, validation, cross-validation, framework structure, DDPG formulation, units, case generality, and broader-audience accessibility. The revised manuscript responds by simplifying the framing, adding definitions and tables, expanding the method explanation, and keeping the claims within the analytic response-generation and limited sensitivity-analysis scope.

■ **Comment R1-1.**

In the Abstract, the paragraph is too long (more than 250 words), which can be simplified. Furthermore, it is recommended that authors point out the significance of the work with more brief but powerful expressions.

Response.

We thank the reviewer for this comment. We agree that the original Abstract was too long and did not state the significance with enough precision. This point matters because the Abstract is the first place where readers decide whether the paper is a broad optimizer-superiority claim or a bounded surrogate-conditioned benchmarking study.

We revised the Abstract, final Introduction contribution paragraph, and Conclusions to reorder the abstract around the block-scale design problem, generated descriptors, analytic response surface, selected 2000-sample DNN surrogate, optimizer comparison, feasible projection, 24-case cross-model stress test, and four-block climate sensitivity analysis. The revised material now gives the reader a more direct account of what was changed and how the evidence should be read.

The evidence now reports repeated-validation nMAE values of 0.0176, 0.0265, and 0.0145 for EUI_t, EG, and H; NSGA-II near-ceiling descriptor-space HV under the fixed reference; compression of the 2000-candidate NSGA-II archive to 51 unique feasible blocks after projection; weak physical metric and rank agreement; and climate-dependent EUI_t and H responses. The revised wording therefore keeps the significance visible while avoiding a claim of broad physical optimizer superiority.

Revisions made in the manuscript.

We made three corresponding revisions. First, we rewrote the Abstract around the problem, method, key numerical results, and interpretation limits. Second, we revised the final Introduction contribution paragraph so that the significance is framed as representation-aware design support. Third, we revised the Conclusions so that the final message emphasizes projection distance, model agreement, and climate sensitivity rather than direct physical optimizer superiority.

Relevant revised manuscript and supporting evidence.

Main manuscript, Abstract.

Block-scale urban energy design requires optimization methods that connect urban form with building energy demand, rooftop photovoltaic potential, and winter solar access during early-stage planning. This study develops a surrogate-conditioned framework to compare policy learning and Pareto search for residential-block morphology optimization. Feasible block layouts are generated first and converted into 12 morphology descriptors. A morphology-to-performance response surface then assigns three targets to each sample: Energy Use Intensity, Energy Generation, and winter sunlight hours. These targets are denoted EUIt, EG, and H throughout the paper. The 2000-sample DNN surrogate selected from nested 500/1000/1500/2000-sample pools provides the common environment for DDPG and NSGA-II. Repeated five-fold validation gives mean nMAE values of 0.0176, 0.0265, and 0.0145 for EUIt, EG, and H. Under the shared surrogate and matched query budget, NSGA-II attains near-ceiling descriptor-space HV, while DDPG shows preference-dependent performance. Projection to generated feasible blocks compresses the 2000-candidate NSGA-II archive to 51 unique feasible blocks, indicating that descriptor-space coverage and feasible-morphology diversity differ substantially. A 24-case cross-model stress test completes all locked cases and shows limited metric and rank consistency between analytic and physics-based outputs. A four-block cross-climate sensitivity analysis shows stable EG ordering but climate-dependent EUIt and H responses. The study contributes a representation-aware urban design workflow that links surrogate search, feasible-block projection, and sensitivity analysis to improve the interpretability of optimizer comparisons for early-stage urban energy planning.

■ **Comment R1-2.**

Overall, there are too many acronyms which decreases the paper's readability. Moreover, symbols should be added.

Response.

We thank the reviewer for this comment. We agree that the previous front-matter list addressed abbreviations but did not sufficiently support the mathematical notation used in the methodology and optimization equations. This point matters because a reader needs both acronym definitions and symbol definitions before reading the analytic response and DDPG reward equations.

We revised the front matter and Table 1 to replace the Abbreviations block with a Nomenclature block containing Abbreviations and Formula symbols subsections. The revised material now gives the reader a more direct account of what was changed and how the evidence should be read.

The evidence now defines EUIt, EG, H, DNN, DDPG, NSGA-II, HV, IGD, nMAE, OSLI, the morphology descriptor vector, analytic target vector, normalized target coordinate, target bounds, ideal vector, axis-scaling vector, weighted distance, DDPG reward, orientation angle, response-generation noise term, and geometric quantities used by the descriptors. We kept full descriptor construction in Table 1 so that the Nomenclature supports mathematical readability without duplicating the descriptor table.

Revisions made in the manuscript.

We made three corresponding revisions. First, we replaced the front-matter Abbreviations block with a Nomenclature block. Second, we added a Formula symbols subsection for the analytic-response and DDPG reward notation. Third, we retained the detailed construction of the 12 morphology descriptors in Table 1 and added a cross-reference from Nomenclature to Table 1.

Relevant revised manuscript and supporting evidence.

Main manuscript, Nomenclature excerpt.

Nomenclature now contains two subsections. *Abbreviations* defines EUI, EG, H, DNN, DDPG, NSGA-II, CMA-ES, GHI, HV, IGD, nMAE, and OSLI. *Formula symbols* defines the morphology descriptor vector $\mathbf{x}$, analytic target vector $\mathbf{y}^{ana}$, target value y_i , normalized target coordinate z_i , target bounds $y_i^{\min}$ and $y_i^{\max}$, ideal vector $\mathbf{u}$, axis-scaling vector $\mathbf{w}$, weighted distance d_w , DDPG reward R , orientation angle θ , response-generation noise term ϵ , and the geometric area, height, and canyon-width quantities used by the descriptor definitions.

Main manuscript, Nomenclature-to-Table 1 cross-reference.

The morphology descriptor symbols FAR, SD, AF, AR_{ew} , AR_{ns} , SVF, BD, OSR, SC, PAR, θ , and OSLI are defined in Table 1.

Main manuscript, Table 1 excerpt.

Descriptor	Unit or type	Construction	Interpretation
FAR	–	$A_{\text{gross}}/A_{\text{block}}$	density
SD	m	$h_{\text{max}} - \bar{h}$	height variation
AF	floors	$A_{\text{gross}}/A_{\text{foot}}$	linked to FAR, BD
AR_{ew}	–	$\bar{h}/W_{ew}$	east-west canyon
AR_{ns}	–	$\bar{h}/W_{ns}$	north-south canyon
SVF	–	analytic openness proxy	sky-view openness
BD	–	$A_{\text{foot}}/A_{\text{block}}$	footprint density
OSR	–	$A_{\text{open}}/A_{\text{gross}}$	open-space ratio
SC	–	$A_{\text{env}}/A_{\text{gross}}$	envelope exposure
PAR	m^{-1}	$P_{\text{sum}}/A_{\text{foot}}$	compactness
θ	deg	sampled in $[-45, 45]$	orientation
OSLI	integer	0–8	open-space cell

Revised Table 1 excerpt. The 12 quantities are derived morphology descriptors rather than independently sampled design variables.

■ Comment R1-3.

Figure 1: The caption should be improved to explain the content of the figure in more detail.

Response.

We thank the reviewer for this comment. We agree that the original Fig. 1 caption was too brief for a workflow figure that carries much of the methodological structure. This point matters because the figure introduces the distinction between generated block geometries, morphology descriptors, surrogate-based optimizer queries, feasible-block projection, and selected sensitivity analyses.

We revised the methodology transition before Fig. 1 and the Fig. 1 caption to rewrite the transition paragraph and caption so that the workflow stages are introduced in order. The revised material now gives the reader a more direct account of what was changed and how the evidence should be read.

The evidence now identifies feasible morphology generation, analytic response assignment and DNN surrogate training, surrogate-assisted optimization, nearest-feasible-block projection, and limited physical and climate sensitivity analyses. The caption no longer implies that all 2000 generated samples were direct annual EnergyPlus and Radiance simulations.

This placement was deliberate. The workflow figure is now treated as the reader's map for the whole revision, so the response explains not only that the caption was lengthened, but also which methodological misunderstanding the new caption prevents.

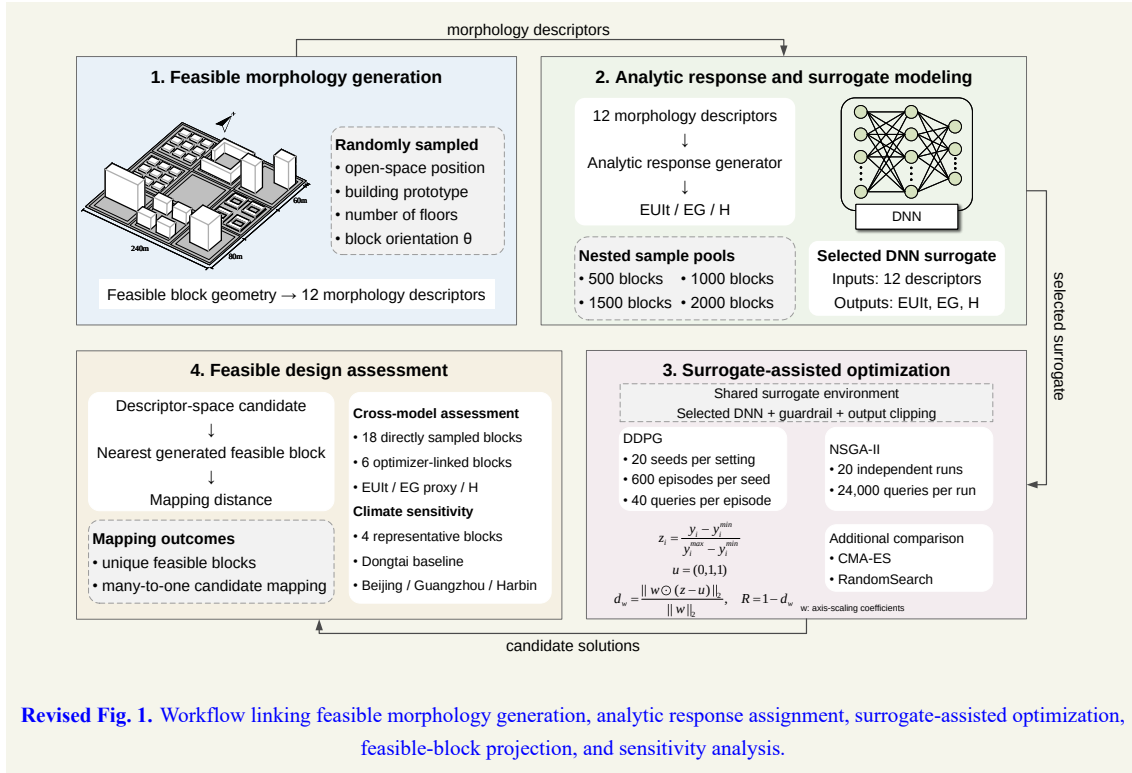
Revisions made in the manuscript.

We made three corresponding revisions. First, we inserted a clearer methodology transition paragraph before Fig. 1. Second, we rewrote the Fig. 1 caption so that each workflow module is explicitly identified. Third, we made the distinction between analytic response generation and selected physical calculations visible before the detailed methodology.

Relevant revised manuscript and supporting evidence.**Main manuscript, methodology transition before Fig. 1.**

The methodology links morphology generation, analytic target assignment, surrogate-assisted optimization, and post-optimization assessment within one block-scale design workflow. The workflow distinguishes three representations: generated feasible block geometries, 12 morphology descriptors used by the surrogate and optimizers, and locked block cases examined through physics-based and climate-sensitive calculations. This distinction is essential because the optimizer operates in descriptor space, whereas design interpretation ultimately depends on generated block geometries. Figure 1 summarizes the computational sequence.

Main manuscript, Fig. 1.



Revised Fig. 1. Workflow linking feasible morphology generation, analytic response assignment, surrogate-assisted optimization, feasible-block projection, and sensitivity analysis.

■ **Comment R1-4.**

Line 36-38: According to the Chinese requires reference. Moreover, the authors need to specify some references for the Grasshopper platform. Please describe the general context of the softwares (tools) and why that softwares are suitable for your analysis.

Response.

We thank the reviewer for this comment. We agree that the manuscript needed a clearer urban-energy and software context. This point matters because the work should read as a block-scale planning study motivated by urban renewal, PV deployment, and climate-sensitive design decisions rather than a generic optimizer exercise.

We revised Sections 1.1, 1.2, 2.2, and 2.6 to expand the urban morphology, building energy, rooftop PV, winter solar access, surrogate modelling, and simulation-based optimization context. The revised material now gives the reader a more direct account of what was changed and how the evidence should be read.

The evidence now clarifies that EnergyPlus- and Radiance-component calculations are used only for selected locked cases in the cross-model stress test and climate sensitivity analysis. The analytic response surface remains a controlled benchmark layer and is not presented as a substitute for detailed final-design simulation.

Revisions made in the manuscript.

We made three corresponding revisions. First, we expanded the Introduction around Chinese urban renewal, PV-oriented planning, and climate resilience. Second, we revised the methodology wording so

that analytic response generation and selected physical calculations are separated. Third, we added literature support for surrogate-assisted optimization and urban-form energy analysis without presenting the workflow as a full physical design engine.

Relevant revised manuscript and supporting evidence.

Main manuscript, Motivation paragraph.

Block morphology influences energy performance through coupled physical mechanisms. Density and height alter heating and cooling demand by changing envelope exposure, mutual shading, and micro-climatic context. Solar access and daylight availability depend on roof exposure, sky-view conditions, block orientation, and open-space arrangement. Recent large-scale studies further show that urban spatial structure can shape building energy demand across broad urban areas and under extreme climate events, which makes morphology-sensitive design relevant beyond isolated building cases.

Main manuscript, Section 2.2 paragraph.

This module provides a controlled morphology-to-performance response surface for optimizer comparison. The nested 500/1000/1500/2000-sample pools are deterministic subsets of the same generated morphology pool under the recorded random seed and morphology-generation protocol. The physics-based case calculations are conducted subsequently on locked generated blocks to assess how the analytic target values relate to EnergyPlus- and Radiance-component outputs in selected cases.

Supplementary Material, Section S5.3 paragraph.

The 18 direct feasible cases completed both EnergyPlus and Radiance-component runs. EUIt agreement is weak, EG is represented by a GHI-based rooftop-PV proxy, and H is represented by January 20 windowsill direct-sun hours. The nMAE and rank-correlation values therefore define a stress-test boundary for the tested direct cases.

Local references.

- Javanroodi, K., Perera, A. T. D., et al. (2023). Designing climate resilient energy systems in complex urban areas considering urban morphology: A technical review. *Advances in Applied Energy*, 12. <https://doi.org/10.1016/j.adapen.2023.100155>
- Du, L., Wang, H., et al. (2025). Impact of block form on building energy consumption, urban microclimate and solar potential: A case study of Wuhan, China. *Energy and Buildings*, 328. <https://doi.org/10.1016/j.enbuild.2024.115224>
- Wang, H., Wu, Z., Gu, W., et al. (2026). Towards climate-resilient cities: Assessing and planning urban energy demand under extreme climate events. *Energy and Buildings*, 337, 117436. <https://doi.org/10.1016/j.enbuild.2026.117436>

■ Comment R1-5.

Line 49: It would be good to describe why you select 12 parameters for building models.

Response.

We thank the reviewer for this comment. We agree that the descriptor-selection logic and dependency

structure were not explained clearly enough. This point matters because the 12 morphology quantities are derived descriptors from generated block geometries rather than independent variables that can vary freely.

We revised Section 2.1, Table 1, and Supplementary Sections S1.2–S1.5 to rewrite the descriptor explanation, define the descriptors with units and construction notes, and add supporting material on prototype settings, OSLI, dependency, PCA coverage, and distributions. The revised material now gives the reader a more direct account of what was changed and how the evidence should be read.

The evidence now shows that FAR, BD, AF, and OSR share footprint and gross-floor-area quantities, that OSLI is integer-valued by construction, that six principal components explain 95 percent of descriptor variance, and that exact duplicate descriptor rows are absent in the canonical 2000-row pool. Optimizer outputs are therefore interpreted first in descriptor space and then through nearest-feasible-block projection before morphology-level conclusions are drawn.

Revisions made in the manuscript.

We made three corresponding revisions. First, we rewrote the descriptor-selection paragraph around density, height, openness, envelope exposure, compactness, orientation, and open-space position. Second, we revised Table 1 to include unit or type, construction, and interpretation. Third, we added supporting material on descriptor dependency and PCA coverage.

Relevant revised manuscript and supporting evidence.

Main manuscript, Section 2.1 paragraphs.

The morphology generator first constructs feasible block layouts and only then calculates descriptor values. Each block is represented as a 240 m $\times$ 240 m square divided into a 3 $\times$ 3 grid of 80 m $\times$ 80 m land units. One cell is assigned as open space, and the remaining eight cells are populated with residential prototypes. For each generated block, the generator samples the open-space position, building prototype, number of floors, and orientation angle. The orientation angle is sampled within -45 deg to 45 deg. The resulting block is therefore a generated feasible morphology, while the 12 input quantities used by the surrogate are descriptors derived from that morphology.

The descriptor set in Table 1 includes FAR, SD, AF, AR_{ew}, AR_{ns}, SVF, BD, OSR, SC, PAR, theta, and OSLI. These quantities are derived morphology descriptors with geometric dependencies. For example, FAR, BD, AF, and OSR are linked through the same footprint and gross-floor-area quantities, while OSLI is an integer descriptor of open-space position. Consequently, optimizer outputs are first interpreted in descriptor space and then mapped to generated feasible blocks before morphology-level conclusions are drawn.

Main manuscript, Table 1 excerpt.

Descriptor	Unit or type	Construction	Interpretation
FAR	—	$A_{\text{gross}}/A_{\text{block}}$	density
SD	m	$h_{\text{max}} - \bar{h}$	height variation
AF	floors	$A_{\text{gross}}/A_{\text{foot}}$	linked to FAR, BD
AR_{ew}	—	$\bar{h}/W_{ew}$	east-west canyon
AR_{ns}	—	$\bar{h}/W_{ns}$	north-south canyon
SVF	—	analytic openness proxy	sky-view openness
BD	—	$A_{\text{foot}}/A_{\text{block}}$	footprint density
OSR	—	$A_{\text{open}}/A_{\text{gross}}$	open-space ratio
SC	—	$A_{\text{env}}/A_{\text{gross}}$	envelope exposure
PAR	m^{-1}	$P_{\text{sum}}/A_{\text{foot}}$	compactness
θ	deg	sampled in $[-45, 45]$	orientation
OSLI	integer	0–8	open-space cell

Revised Table 1 excerpt. The 12 quantities are derived morphology descriptors rather than independently sampled design variables.

Supplementary Material, Table S2 excerpt.

Check	Value	Interpretation
FAR minus BD times AF	1.33×10^{-15}	numerical roundoff
OSR relation residual	2.22×10^{-16}	footprint relation holds
OSLI integer consistency	1.0	all OSLI values are integer
Exact duplicate count	0	no exact duplicate descriptor rows
PCA components for 95% variance	6	descriptor space is structured

Supplementary Material, Fig. S1.

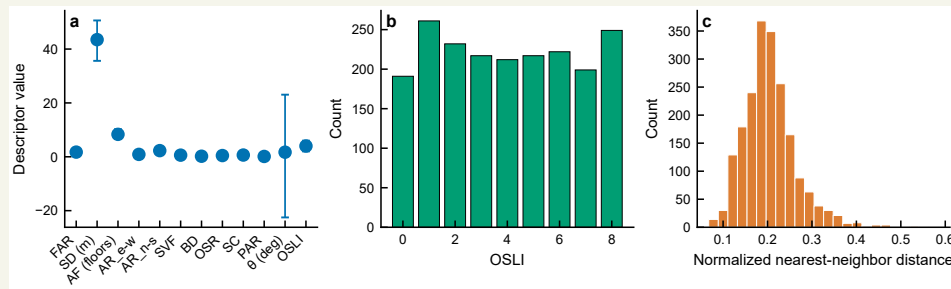


Fig. S1 excerpt. Descriptor distributions and nearest-neighbour distances in the generated morphology pool.

Comment R1-6.

Table 1: The author could add an additional column on the right to describe Factors (Floor area ratio, Setback, diversity, etc.). Table 2: The parameters are not assumptions; it seems the baseline or operating parameters for the building energy models.

Response.

We thank the reviewer for this comment. We agree that the functions of Table 1 and Table 2 were not sufficiently distinct. This point matters because readers need to know which table defines morphology inputs and which table defines evaluation modes.

We revised Tables 1 and 2 and Sections 2.1–2.2 to separate descriptor construction from evaluation-mode interpretation. The revised material now gives the reader a more direct account of what was changed and how the evidence should be read.

The evidence now Table 1 now defines morphology descriptors by unit or type, construction, and interpretation, while Table 2 distinguishes analytic response generation, surrogate prediction, feasible projection, physical stress testing, and climate sensitivity by source, output, and boundary. This split avoids mixing descriptor construction, target assignment, surrogate prediction, and selected physical calculations in one undifferentiated method description.

This table separation also makes later evidence easier to follow because response-surface construction, surrogate prediction, and physical stress testing now have different table anchors.

Revisions made in the manuscript.

We made three corresponding revisions. First, we rewrote Table 1 as the descriptor-definition table. Second, we rewrote Table 2 as the evaluation-mode table. Third, we aligned the surrounding method text with the two-table division.

Relevant revised manuscript and supporting evidence.

Main manuscript, Table 1 excerpt.

Descriptor	Unit or type	Construction	Interpretation
FAR	–	$A_{\text{gross}}/A_{\text{block}}$	density
SD	m	$h_{\text{max}} - \bar{h}$	height variation
AF	floors	$A_{\text{gross}}/A_{\text{foot}}$	linked to FAR, BD
AR_{ew}	–	$\bar{h}/W_{ew}$	east-west canyon
AR_{ns}	–	$\bar{h}/W_{ns}$	north-south canyon
SVF	–	analytic openness proxy	sky-view openness
BD	–	$A_{\text{foot}}/A_{\text{block}}$	footprint density
OSR	–	$A_{\text{open}}/A_{\text{gross}}$	open-space ratio
SC	–	$A_{\text{env}}/A_{\text{gross}}$	envelope exposure
PAR	m^{-1}	$P_{\text{sum}}/A_{\text{foot}}$	compactness
θ	deg	sampled in $[-45, 45]$	orientation
OSLI	integer	0–8	open-space cell

Revised Table 1 excerpt. The 12 quantities are derived morphology descriptors rather than independently sampled design variables.

Main manuscript, Table 2 excerpt.

Mode	Source	Output	Boundary
Analytic response generation	generated descriptors	EUI _t , EG, H	analytic surface
Surrogate prediction	12 descriptors	DNN outputs	selected checkpoint
Feasible projection	descriptor candidate	nearest block	nearest neighbour
Physical stress test	locked blocks	physical outputs	24 cases
Climate sensitivity	four blocks	weather deltas	no retraining

Revised Table 2 excerpt. Evaluation modes are separated by source, output, and interpretation boundary.

■ Comment R1-7.

The author may need to justify the methodology used. I think it would be great to have model validation studies using published literature data or exp. data. Moreover, the authors need to describe the underlying assumptions of their models.

Response.

We thank the reviewer for this comment. We agree that the methodological assumptions and validation hierarchy were not visible enough. This point matters because surrogate accuracy on analytic targets cannot be treated as physical equivalence or optimizer-ranking transfer.

We revised Sections 2.2, 2.3, 2.6, 3.5, and 4.2, Tables 3 and 6, Fig. 5, Fig. 9, and Supplementary Sections S1, S2, and S5 to separate analytic response generation, DNN surrogate validation against analytic targets, feasible projection, and limited physics-based cross-model stress testing. The revised material now gives the reader a more direct account of what was changed and how the evidence should be read.

The evidence now reports repeated 5 by 5 validation nMAE values of 0.0176, 0.0265, and 0.0145 for EUI_t, EG, and H, then reports weak physical agreement in the 18 direct feasible cases with Spearman rho values of -0.404 for EUI_t, -0.023 for the EG proxy, and -0.043 for H. The physical calculations are therefore interpreted as a stress test showing fragility, not as validation of direct physical optimizer superiority.

Revisions made in the manuscript.

We made four corresponding revisions. First, we added the analytic-response equations, stochastic terms, and clipping ranges to the supporting material. Second, we expanded the surrogate-validation section to report repeated and boundary-oriented checks. Third, we added the 24-case physical stress-test summary. Fourth, we rewrote the Discussion so that weak physical agreement limits descriptor-space optimizer interpretation.

Relevant revised manuscript and supporting evidence.

Main manuscript, Section 2.2 paragraph.

In this study, the analytic response generator is the target-assignment module that maps a generated morphology descriptor vector, denoted $\mathbf{x}$, to three target responses, denoted $\mathbf{y}^{ana} = [\text{EUI}_t, \text{EG}, H]^T$. The mapping uses explicit formulas that combine density, openness, aspect ratio, sky-view factor, orientation,

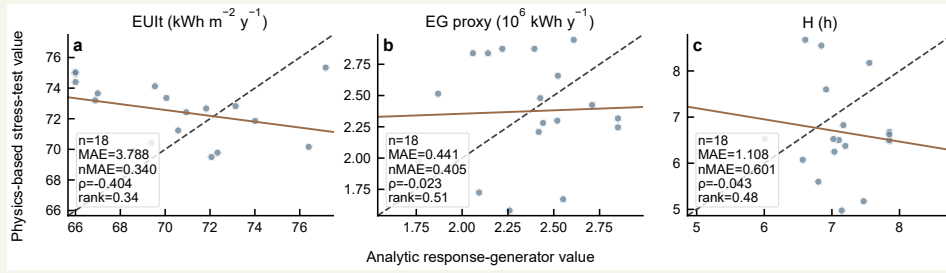
and open-space position to represent demand, rooftop-generation, and sunlight-access responses. A small stochastic term is added during dataset construction, and the target values are then limited to the response ranges used for surrogate training.

Main manuscript, Table 3 excerpt.

Validation regime	EUI _t nMAE	EG nMAE	H nMAE	EUI _t ρ	EG ρ	H ρ
Repeated 5×5 CV	0.0176	0.0265	0.0145	0.991	0.984	0.995
Leave-one-OSLI-out	0.0185	0.0342	0.0185	0.990	0.972	0.992
Outer-shell holdout	0.0278	0.0335	0.0208	0.985	0.975	0.994
Feature-tail holdout	0.0230	0.0287	0.0198	0.986	0.983	0.992

Revised Table 3 excerpt. nMAE and Spearman ρ compare DNN predictions with analytic target values.

Main manuscript, Fig. 9.



Revised Fig. 9. Cross-model stress test for 18 direct generated feasible blocks.

Supplementary Material, Table S4 excerpt.

Target	Count	MAE	nMAE	Spearman ρ
EUI _t	18	3.788	0.340	-0.404
EG GHI proxy	18	0.441	0.405	-0.023
H	18	1.108	0.601	-0.043

Revised Table 6 and Table S4 excerpt. The direct feasible stress-test cases show limited metric and rank consistency.

■ Comment R1-8.

The author may need to describe what the five-fold cross validation means.

Response.

We thank the reviewer for this comment. We agree that the cross-validation explanation should be understandable beyond the metric values. This point matters because small error numbers are not enough

unless the reader knows what was held out and what evidence source was being tested.

We revised Section 2.3, Table 3, Fig. 5, and Supplementary Sections S2.2–S2.5 to explain repeated five-fold cross-validation as repeated held-out testing of the selected 2000-sample DNN surrogate against analytic target values. The revised material now gives the reader a more direct account of what was changed and how the evidence should be read.

The evidence now keeps the repeated nMAE and Spearman metrics and adds OSLI holdout, outer-shell holdout, and feature-tail holdout checks for more difficult descriptor regions. The evidence assesses DNN agreement with analytic targets, not agreement with EnergyPlus or Radiance outputs.

Revisions made in the manuscript.

We made three corresponding revisions. First, we rewrote the cross-validation explanation in plain methodological terms. Second, we retained the repeated 5 by 5 nMAE and Spearman values in Table 3. Third, we added boundary-oriented validation families in the supporting material.

Relevant revised manuscript and supporting evidence.

Main manuscript, Section 2.3 paragraph.

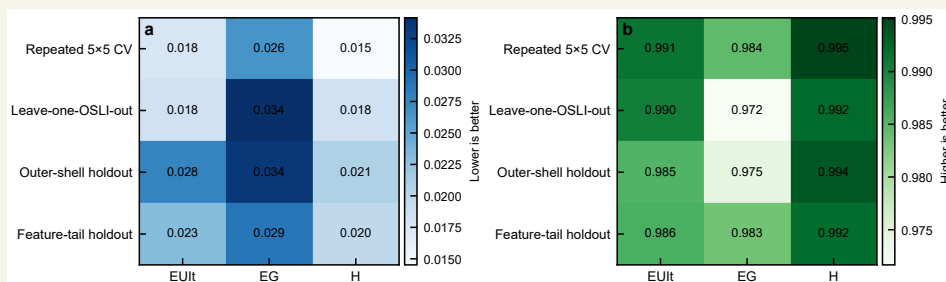
The selected surrogate is assessed through repeated five-fold cross-validation, leave-one-OSLI-out validation, outer-shell holdout, and feature-tail holdout. The repeated five-fold protocol evaluates interpolation performance under randomized folds. The leave-one-OSLI-out protocol tests whether the surrogate remains stable when one open-space-position class is withheld. The outer-shell and feature-tail holdouts examine surrogate behaviour near the boundary of the sampled descriptor domain.

Main manuscript, Table 3 excerpt.

Validation regime	EUIt nMAE	EG nMAE	H nMAE	EUIt ρ	EG ρ	H ρ
Repeated 5×5 CV	0.0176	0.0265	0.0145	0.991	0.984	0.995
Leave-one-OSLI-out	0.0185	0.0342	0.0185	0.990	0.972	0.992
Outer-shell holdout	0.0278	0.0335	0.0208	0.985	0.975	0.994
Feature-tail holdout	0.0230	0.0287	0.0198	0.986	0.983	0.992

Revised Table 3 excerpt. nMAE and Spearman ρ compare DNN predictions with analytic target values.

Main manuscript, Fig. 5.



Revised Fig. 5. Surrogate assessment across random, grouped, and boundary-oriented validation regimes.

Supplementary Material, Section S2.2 paragraph.

Repeated five-fold cross-validation gives mean nMAE values of 0.0176 for EUI, 0.0265 for EG, and 0.0145 for H. The corresponding Spearman correlations are 0.991, 0.984, and 0.995. These values support low error within the analytic response-generation regime.

■ **Comment R1-9.**

The framework has two components: The first The second ...

Response.

We thank the reviewer for this comment. We agree that the framework components needed to be introduced before the detailed results. This point matters because the workflow is easier to audit when each result is tied to the representation and evaluation mode that produced it.

We revised the Methodology opening, Fig. 1, Table 2, and Section 2.6 to rewrite the workflow as a sequence of feasible morphology generation, descriptor extraction, analytic response assignment, surrogate training, optimizer querying, projection, and selected sensitivity analyses. The revised material now gives the reader a more direct account of what was changed and how the evidence should be read.

The evidence now lets the reader trace whether a result comes from the analytic benchmark, the selected DNN surrogate, feasible projection, physical stress testing, or climate sensitivity. The full 2000-sample benchmark is now explicitly described as analytic response assignment plus surrogate prediction, not a set of direct annual physical simulations.

Revisions made in the manuscript.

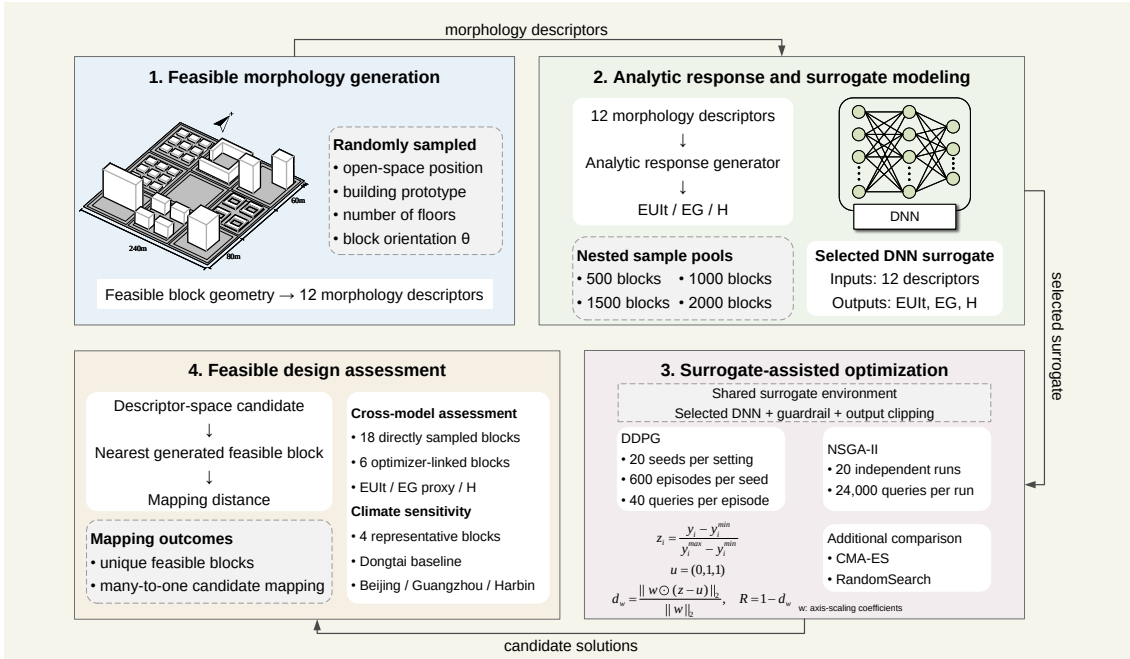
We made three corresponding revisions. First, we rewrote the opening methodology paragraph around workflow components. Second, we revised Fig. 1 and its caption to show the same component order. Third, we used Table 2 to define each evaluation mode and its boundary.

Relevant revised manuscript and supporting evidence.

Main manuscript, methodology opening paragraph.

The methodology links morphology generation, analytic target assignment, surrogate-assisted optimization, and post-optimization assessment within one block-scale design workflow. The workflow distinguishes three representations: generated feasible block geometries, 12 morphology descriptors used by the surrogate and optimizers, and locked block cases examined through physics-based and climate-sensitive calculations.

Main manuscript, Fig. 1.



Revised Fig. 1. Workflow linking feasible morphology generation, analytic response assignment, surrogate-assisted optimization, feasible-block projection, and sensitivity analysis.

Main manuscript, Table 2 excerpt.

Mode	Source	Output	Boundary
Analytic response generation	generated descriptors	EUI, EG, H	analytic surface
Surrogate prediction	12 descriptors	DNN outputs	selected checkpoint
Feasible projection	descriptor candidate	nearest block	nearest neighbour
Physical stress test	locked blocks	physical outputs	24 cases
Climate sensitivity	four blocks	weather deltas	no retraining

Revised Table 2 excerpt. Evaluation modes are separated by source, output, and interpretation boundary.

Comment R1-10.

Please add a reference for Markov Decision Process.

Response.

We thank the reviewer for this comment. We agree that the reinforcement-learning formulation and references needed more conservative framing. This point matters because the static design-search problem should not be overstated as a natural time-evolving MDP.

We revised Sections 1.2 and 2.4 and Fig. 3 to add RL/DDPG references and rewrote DDPG as a policy-learning comparator in a bounded surrogate-query environment. The revised material now gives the reader a more direct account of what was changed and how the evidence should be read.

The evidence now defines state as a serialized descriptor vector, action as a descriptor-space candidate

proposal, reward as normalized weighted target distance, and an episode as a fixed sequence of surrogate queries. DDPG is therefore used as a continuous-action query generator under the fixed surrogate, not as a model of physical urban evolution.

This conservative wording keeps the RL reference useful while making clear that the present benchmark is a static design-query setting, not a physical control process.

Revisions made in the manuscript.

We made three corresponding revisions. First, we added reinforcement-learning and DDPG references. Second, we rewrote the DDPG formulation around state, action, reward, episode length, actor, critic, and target networks. Third, we softened the wording so that the manuscript does not overstate the task as a natural MDP.

Relevant revised manuscript and supporting evidence.

Main manuscript, Section 1.2 paragraph.

Deep reinforcement learning provides another family of search methods, but its role in morphology design requires careful framing. General DRL reviews and foundational DQN and DDPG studies establish policy learning for sequential decision-making and continuous control. In the present study, DDPG is adopted as a continuous-action policy-learning comparator because it provides a clear actor-critic mechanism for generating descriptor queries in a bounded surrogate environment.

Main manuscript, Section 2.4 paragraph.

The DDPG environment serializes repeated queries to a static guarded surrogate. At the beginning of each episode, a random descriptor query is evaluated to construct the initial normalized three-target state. At each subsequent step, the actor outputs an absolute 12-dimensional normalized descriptor action. This action is not an incremental modification of the preceding block. The action is denormalized to descriptor bounds, evaluated by the guarded surrogate, and converted to the next normalized target state and scalar reward. The 40-step horizon therefore functions as a fixed sequence of policy-training queries on a static surrogate response surface.

Main manuscript, reward formula excerpt.

$$z_i = \frac{y_i - y_i^{\min}}{y_i^{\max} - y_i^{\min}},$$

$$\mathbf{u} = [0, 1, 1]^T.$$

$$d_w = \frac{\|\mathbf{w} \odot (\mathbf{z} - \mathbf{u})\|_2}{\|\mathbf{w}\|_2}, \quad R = 1 - d_w.$$

Local references.

Wang, X., Wang, S., et al. (2024). Deep reinforcement learning: A survey. *IEEE Transactions on Neural Networks and Learning*

- Systems*, 35(4), 5064–5078. <https://doi.org/10.1109/tnnls.2022.3207346>
- Mnih, V., Kavukcuoglu, K., et al. (2015). Human-level control through deep reinforcement learning. *Nature*, 518(7540), 529–533. <https://doi.org/10.1038/nature14236>
- Lillicrap, T. P., Hunt, J. J., et al. (2015). Continuous control with deep reinforcement learning. *arXiv preprint arXiv:1509.02971*. <https://doi.org/10.48550/arXiv.1509.02971>

■ Comment R1-11.

Figure 3: The manuscript does not include the explanation of the Figure 3. Moreover, the authors need to explain the symbols and equations in the Figure 3.

Response.

We thank the reviewer for this comment. We agree that the DDPG symbols and equations were too implicit. This point matters because readers need the actor-critic components and reward path to understand how DDPG queries the surrogate benchmark.

We revised Section 2.4 and Fig. 3 to define the state, action, reward, actor, critic, replay buffer, and target networks and redraw Fig. 3 around this architecture. The revised material now gives the reader a more direct account of what was changed and how the evidence should be read.

The evidence now shows target normalization, the ideal vector $[0, 1, 1]^T$, the weighted normalized distance d_w , and the corrected reward $R = 1 - d_w$. The formulation explains query scoring in the selected surrogate environment without implying that DDPG learns physical building behaviour.

The response therefore links the corrected equations to the architecture figure, so readers can trace how a descriptor query becomes a surrogate prediction and then a scalar reward.

Revisions made in the manuscript.

We made three corresponding revisions. First, we expanded Section 2.4 to define the DDPG components. Second, we redrew Fig. 3 as a static surrogate-query architecture diagram. Third, we added the corrected normalization and reward equations.

Relevant revised manuscript and supporting evidence.

Main manuscript, Section 2.4 paragraph defining state/action/episode.

The DDPG environment serializes repeated queries to a static guarded surrogate. At the beginning of each episode, a random descriptor query is evaluated to construct the initial normalized three-target state. At each subsequent step, the actor outputs an absolute 12-dimensional normalized descriptor action. This action is not an incremental modification of the preceding block. The action is denormalized to descriptor bounds, evaluated by the guarded surrogate, and converted to the next normalized target state and scalar reward.

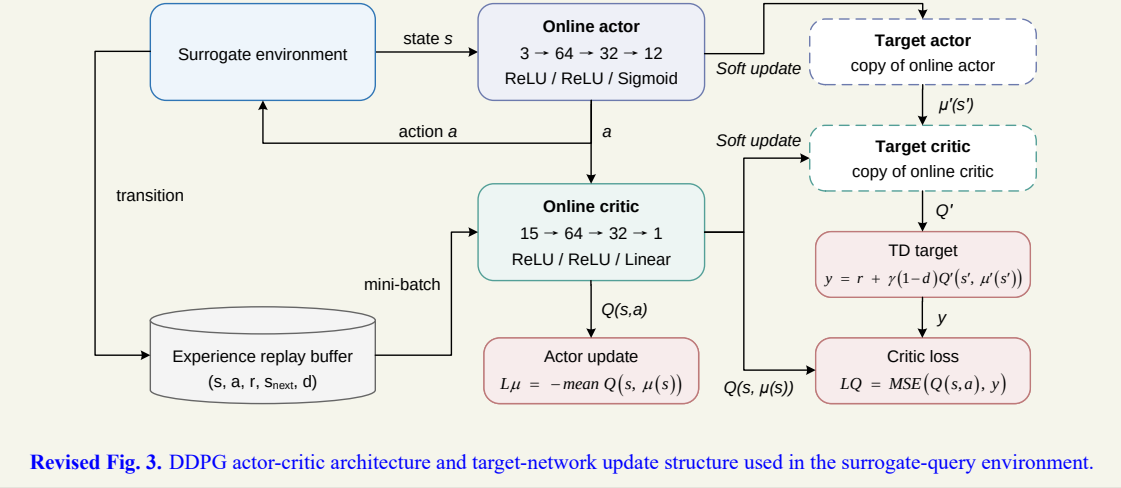
Main manuscript, reward equations.

$$z_i = \frac{y_i - y_i^{\min}}{y_i^{\max} - y_i^{\min}}.$$

$$\mathbf{u} = [0, 1, 1]^T.$$

$$d_w = \frac{\|\mathbf{w} \odot (\mathbf{z} - \mathbf{u})\|_2}{\|\mathbf{w}\|_2}, \quad R = 1 - d_w.$$

Main manuscript, Fig. 3.



■ **Comment R1-12.**

Figure 4: For parity plots, the ticks and limits of the x- and y-axes should be same. All plots should be rescaled. Please specify unit in each axis.

Response.

We thank the reviewer for this comment. We agree that the figure and table units needed clearer labelling. This point matters because unit ambiguity makes it hard to distinguish raw targets, normalized coordinates, and morphology descriptors.

We revised Figs. 4, 5, and 9, Tables 1, 3, and 6, and Supplementary captions and tables to standardize target units, morphology units, one-to-one reference lines, and figure captions. The revised material now gives the reader a more direct account of what was changed and how the evidence should be read.

The evidence now labels EUIt in $\text{kWh m}^{-2} \text{y}^{-1}$, EG in 10^6kWh y^{-1} where that scale is used, H in hours, and morphology quantities by their unit or type. This correction changes presentation only; the underlying numerical results are unchanged.

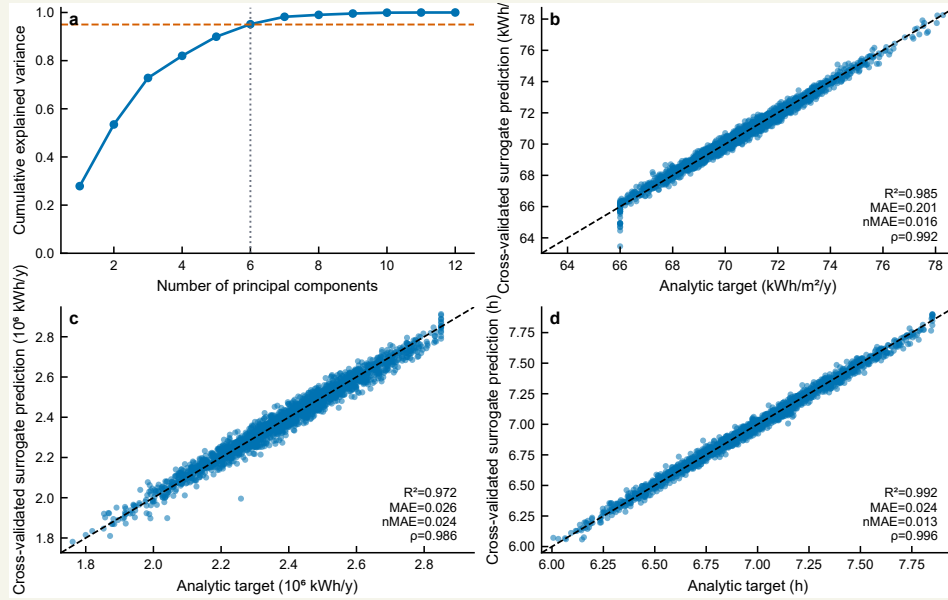
The response also distinguishes unit correction from result revision, which is important because the numbers are retained while the labels and comparison guides become easier to audit.

Revisions made in the manuscript.

We made three corresponding revisions. First, we standardized target-unit labels in the relevant figures and tables. Second, we revised comparison plots to include one-to-one reference lines where appropriate. Third, we checked Supplementary Information captions and tables against the same convention.

Relevant revised manuscript and supporting evidence.

Main manuscript, Fig. 4.



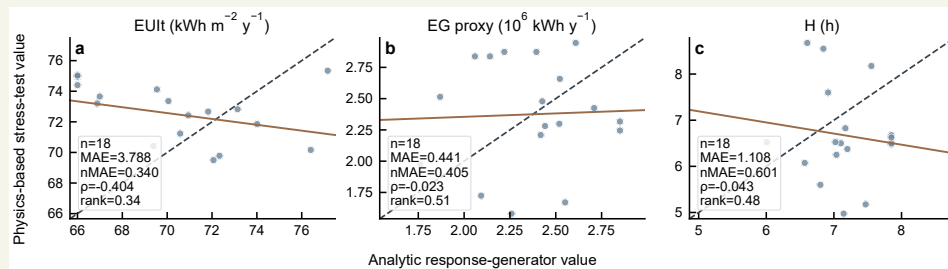
Revised Fig. 4. Descriptor coverage and cross-validated surrogate predictions.

Main manuscript, Table 3 excerpt with units in target definitions.

Validation regime	EUI _t nMAE	EG nMAE	H nMAE	EUI _t ρ	EG ρ	H ρ
Repeated 5×5 CV	0.0176	0.0265	0.0145	0.991	0.984	0.995
Leave-one-OSLI-out	0.0185	0.0342	0.0185	0.990	0.972	0.992
Outer-shell holdout	0.0278	0.0335	0.0208	0.985	0.975	0.994
Feature-tail holdout	0.0230	0.0287	0.0198	0.986	0.983	0.992

Revised Table 3 excerpt. nMAE and Spearman ρ compare DNN predictions with analytic target values.

Main manuscript, Fig. 9.



Revised Fig. 9. Cross-model stress test for 18 direct generated feasible blocks.

■ Comment R1-13.

Figure 6: Please explain what the episode means in the plots.

Response.

We thank the reviewer for this comment. We agree that the DDPG episode definition could be mistaken for physical time evolution. This point matters because in this static surrogate benchmark an episode is a query sequence rather than an annual or dynamic building-operation process.

We revised Sections 2.4 and 3.2, Figs. 2 and 6, and Supplementary Section S3.1 to define one episode as 40 surrogate-query steps and align the training-curve captions with that definition. The revised material now gives the reader a more direct account of what was changed and how the evidence should be read.

The evidence now makes the reward at each step depend on the surrogate-predicted normalized target vector and preference weighting. The episode axis therefore describes training progress in descriptor-query space, not a physical calendar or dynamic urban system.

This explanation is included because the word episode is familiar from sequential control, but here it only denotes the fixed number of surrogate interactions used for training.

Revisions made in the manuscript.

We made three corresponding revisions. First, we rewrote the episode definition in Section 2.4. Second, we revised Fig. 2 and Fig. 6 captions around surrogate-query steps. Third, we added supporting detail on seed-level DDPG behaviour in Supplementary Section S3.1.

Relevant revised manuscript and supporting evidence.

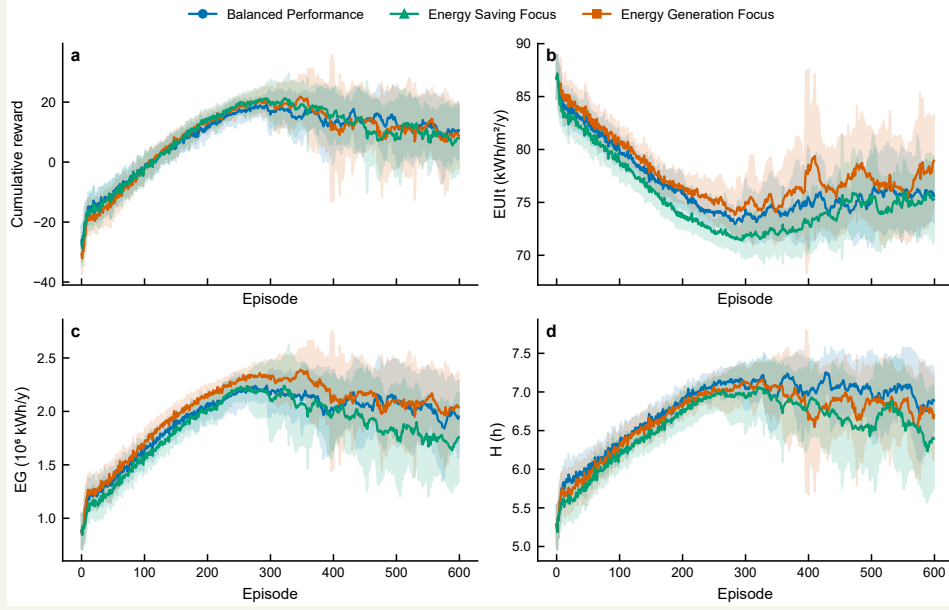
Main manuscript, Section 2.4 paragraph.

The 40-step horizon therefore functions as a fixed sequence of policy-training queries on a static surrogate response surface.

Main manuscript, Section 3.2 paragraphs.

Figure 6 describes how the DDPG agents interact with the guarded surrogate during training. Across the three preference scenarios, rewards increase rapidly during early training and then enter a high-reward region with visible seed-to-seed variability. Because each episode contains a fixed sequence of 40 surrogate queries, the trajectories represent policy-training behaviour on a static response surface. They should therefore be read as search trajectories over descriptor queries, not as the physical evolution of one block design.

The later portions of the curves remain informative. The agents can repeatedly reach favourable surrogate regions, yet the best-to-final gap indicates that late-stage regression remains present. This behaviour is consistent with the retained-output strategy used in Table 4, where one best scalarized candidate is retained from each seed.



Revised Fig. 6. DDPG training trajectories across the three preference scenarios.

Supplementary Material, Section S3.1 paragraph.

Each DDPG scenario uses 20 seeds, 600 episodes, and 40 surrogate interactions per episode. The seed-level summaries record best reward, final reward, plateau episode, and best-to-final regression. These quantities describe policy-training behaviour under serialized surrogate queries on a static surrogate response surface.

■ **Comment R1-14.**

It would be better to conduct more generalized case studies, including different locations (different weather conditions in China) or different building types (an energy-efficient building, an old building, etc.).

Response.

We thank the reviewer for this comment. We agree that the generality of the selected cases and climate analysis needed clearer limits. This point matters because the benchmark dataset and the smaller physical or climate-sensitive case set answer different questions.

We revised Sections 2.6, 3.5, 4.2, and 4.3, Fig. 10, Table 6, and Supplementary Section S6 to add a four-block by three-climate sensitivity analysis and state that building types and climate-aware surrogate retraining remain outside the current scope. The revised material now gives the reader a more direct account of what was changed and how the evidence should be read.

The evidence now shows relatively stable EG ordering in the four-block set but climate-dependent EUI

and H responses, especially under severe-cold conditions. The analysis is a bounded sensitivity check for selected locked cases, not nationwide climate-zone generalization.

Revisions made in the manuscript.

We made three corresponding revisions. First, we defined the four-block by three-climate sensitivity design. Second, we added Fig. 10, Table 6, and Supplementary Section S6 material. Third, we revised the Discussion to identify building-type expansion and climate-aware surrogate training as future work.

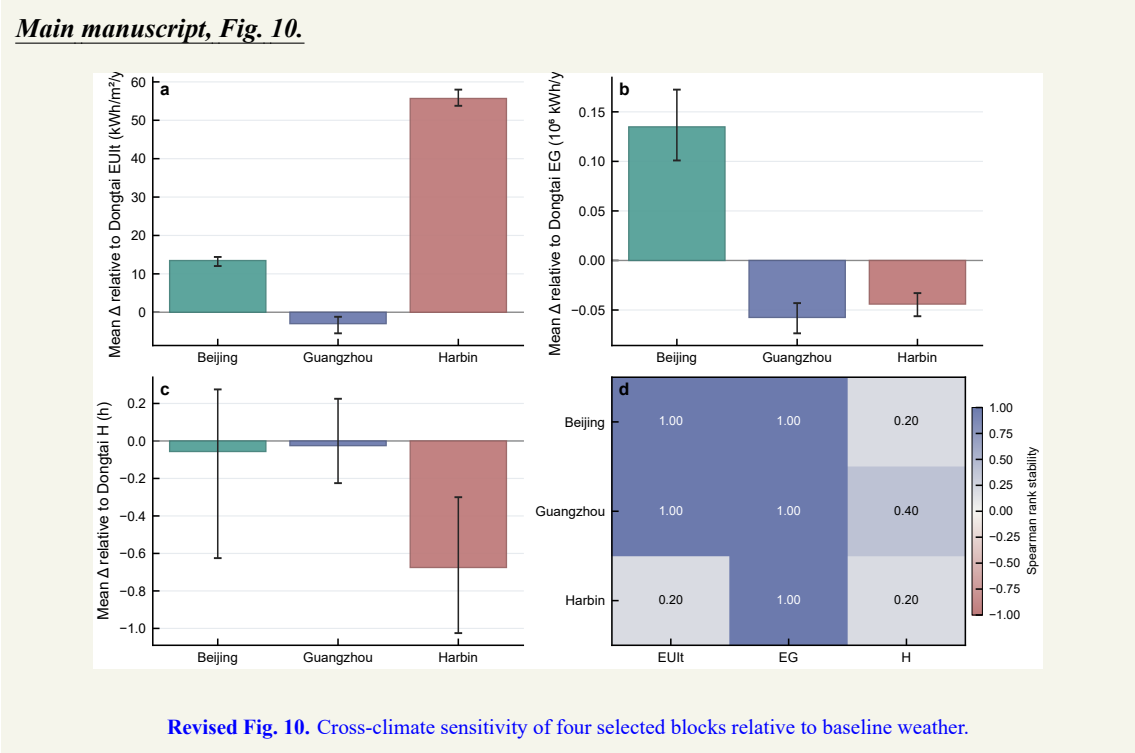
Relevant revised manuscript and supporting evidence.

Main manuscript, Section 2.6 paragraph.

The baseline weather file used in the physical workflow serves as the reference condition for the cross-climate sensitivity analysis. The cross-climate sensitivity set then evaluates the same four block morphologies under three additional weather files representing Harbin, Beijing, and Guangzhou. The surrogate is not retrained, and the morphology set is not reselected after observing climate results.

Main manuscript, Section 3.5 paragraph.

The climate analysis in Fig. 10 evaluates four selected blocks under three additional weather files relative to the baseline weather. The demand response is strongly climate-dependent, especially under the severe-cold Harbin weather. EG ordering remains stable across the additional climates, suggesting that the rooftop-generation proxy is less sensitive to the weather substitution for this four-block set. H ordering is less stable, which is consistent with its dependence on solar geometry, obstruction, and weather-specific direct-sun conditions.



Supplementary Material, Table S5 excerpt.

Station	ΔEUI_t	ΔEG	ΔH	Main interpretation
Beijing	13.467	0.135	-0.056	higher demand
Guangzhou	-2.972	-0.057	-0.025	lower demand
Harbin	55.681	-0.044	-0.675	strong demand increase

Revised Table S5 excerpt. ΔEUI_t is in $\text{kWh m}^{-2} \text{y}^{-1}$, ΔEG is in 10^6kWh y^{-1} , and ΔH is in h.

Supplementary Material, Section S6.3 paragraph.

The rank-stability results indicate target-specific climate sensitivity. EG has stable ordering across the three additional climates, with Spearman ρ equal to 1.0 in each case. H shows lower ordering stability, with Spearman ρ values of 0.2 for Beijing, 0.4 for Guangzhou, and 0.2 for Harbin. EUI_t ordering remains stable for Beijing and Guangzhou but changes under Harbin, where Spearman ρ is 0.2. These results support a limited climate-sensitivity interpretation for the selected four-block set.

■ **Comment R1-15.**

I think it would be great if the authors could write this paper for a broader audience.

Response.

We thank the reviewer for this comment. We agree that the manuscript needed to be easier for Applied Energy readers to follow. This point matters because the original version introduced acronyms, workflow stages, optimizer terminology, and validation language too densely.

We revised the Abstract, Nomenclature, Introduction, Table 2, Discussion, and Supplementary Information structure to rewrite the reader entry points, add Nomenclature, separate descriptor definitions from evaluation modes, and integrate limitations into the Discussion. The revised material now gives the reader a more direct account of what was changed and how the evidence should be read.

The evidence now presents the problem, response surface, surrogate, optimizer comparison, projection, stress-test, and climate-sensitivity boundary in a more linear sequence. The paper remains a surrogate-assisted benchmarking and design-support study rather than a broad DRL-superiority claim.

These readability changes also support the reviewer response itself: the same terminology now appears in the front matter, tables, method sections, and evidence excerpts, reducing the chance that readers encounter different labels for the same evidence layer.

Revisions made in the manuscript.

We made four corresponding revisions. First, we rewrote the Abstract and Introduction for a clearer reader entry point. Second, we replaced the front-matter abbreviation list with Nomenclature. Third, we separated descriptor and evaluation-mode explanations into Tables 1 and 2. Fourth, we revised the Discus-

sion and Conclusions so that limitations are integrated with the main interpretation.

Relevant revised manuscript and supporting evidence.

Main manuscript, Introduction motivation excerpt.

Urban energy planning is increasingly shaped by the dual requirement of reducing building-sector emissions and maintaining reliable energy services. At the district and block scales, urban morphology controls floor area, envelope exposure, roof availability, obstruction, and solar access, which together affect both energy demand and distributed generation potential.

Main manuscript, contribution paragraph.

This study develops a surrogate-assisted comparison workflow for block-scale urban energy design. The contributions are threefold: the study establishes a morphology-to-performance workflow in which feasible residential block layouts are generated before 12 morphology descriptors are calculated; the study compares DDPG and NSGA-II under the same selected DNN surrogate and matched query budget; and the study connects surrogate-based optimization with repeated surrogate assessment, feasible-block projection, a 24-case physics-based cross-model stress test, and a four-block cross-climate sensitivity analysis.

Main manuscript, Table 2 excerpt.

Mode	Source	Output	Boundary
Analytic response generation	generated descriptors	EUI _t , EG, H	analytic surface
Surrogate prediction	12 descriptors	DNN outputs	selected checkpoint
Feasible projection	descriptor candidate	nearest block	nearest neighbour
Physical stress test	locked blocks	physical outputs	24 cases
Climate sensitivity	four blocks	weather deltas	no retraining

Revised Table 2 excerpt. Evaluation modes are separated by source, output, and interpretation boundary.

► Reviewer 2.

We thank Reviewer 2 for identifying the central issues around analytic approximation, physical-probe terminology, validation scope, sample generation, RL formulation, evaluation terminology, target formulas, and reproducibility. These comments shaped the largest part of the revision.

■ Comment R2-1.

The manuscript states that the dataset is generated through an "analytic approximation" of the original EnergyPlus/Radiance workflow rather than through direct simulation. Please provide reasons for this workflow.

Response.

We thank the reviewer for this comment. We agree that the reason for using an analytic response dataset needed to be explicit. This point matters because the analytic response surface is the controlled benchmark layer that makes optimizer comparison auditable before selected physical calculations are added.

We revised Sections 2.1–2.3, Table 2, and Supplementary Section S1.4 to define generated feasible morphologies, analytic target assignment, selected DNN surrogate training, and the boundary between analytic and physical evidence. The revised material now gives the reader a more direct account of what was changed and how the evidence should be read.

The evidence now documents the target functions, stochastic terms, clipping ranges, descriptor dependencies, nested sample regime, and common surrogate checkpoint used for DDGP and NSGA-II. The analytic response dataset is not presented as direct physical simulation of every candidate.

This also explains why the revision does not simply add more physical terminology. The analytic dataset is useful precisely because it is controlled and reproducible, while the selected physical cases are used later to test where that controlled benchmark becomes fragile.

Revisions made in the manuscript.

We made three corresponding revisions. First, we rewrote Sections 2.1–2.3 to distinguish generated morphologies, analytic target assignment, and surrogate training. Second, we revised Table 2 to name analytic response generation as its own evaluation mode. Third, we added equations, stochastic terms, clipping ranges, and descriptor dependencies in Supplementary Section S1.4.

Relevant revised manuscript and supporting evidence.

Main manuscript, Section 2.2 paragraphs.

This module provides a controlled morphology-to-performance response surface for optimizer comparison. The nested 500/1000/1500/2000-sample pools are deterministic subsets of the same generated morphology pool under the recorded random seed and morphology-generation protocol. The physics-

based case calculations are conducted subsequently on locked generated blocks to assess how the analytic target values relate to EnergyPlus- and Radiance-component outputs in selected cases.

The response directions follow the design objectives: lower EUI_t is preferred, while higher EG and H are preferred. The active 2000-sample pool is the largest nested sample regime used for the selected DNN surrogate. The full equations, station-bias convention, stochastic terms, and clipping ranges are given in Supplementary Section S1. Table 2 distinguishes analytic response assignment, surrogate prediction, feasible projection, and the post-optimization sensitivity analyses.

Main manuscript, Table 2 excerpt.

Mode	Source	Output	Boundary
Analytic response generation	generated descriptors	EUI _t , EG, H	analytic surface
Surrogate prediction	12 descriptors	DNN outputs	selected checkpoint
Feasible projection	descriptor candidate	nearest block	nearest neighbour
Physical stress test	locked blocks	physical outputs	24 cases
Climate sensitivity	four blocks	weather deltas	no retraining

Revised Table 2 excerpt. Evaluation modes are separated by source, output, and interpretation boundary.

Supplementary Material, Section S1.4 opening paragraph.

The analytic response generator is the target-assignment module that maps a generated morphology descriptor vector, denoted $\mathbf{x}$, to three target responses, denoted $\mathbf{y}^{ana} = [\text{EUI}_t, \text{EG}, H]^T$. The mapping uses explicit formulas that combine density, openness, aspect-ratio, sky-view, orientation, and open-space-position terms to represent demand, rooftop-generation, and sunlight-access responses.

■ **Comment R2-2.**

The study repeatedly refers to "physical probe" and "physical-stack probe," but the description suggests that the probe is still simplified and only partially aligned with the original metrics.

Response.

We thank the reviewer for this comment. We agree that the physical-probe terminology was too strong. This point matters because selected physical calculations should test cross-model fragility rather than validate the entire surrogate or optimizer ranking.

We revised Table 2, Sections 2.6 and 3.5, Table 6, Fig. 9, and Supplementary Sections S5.3–S5.4 to replace physical-probe wording with limited physics-based cross-model stress test and rewrote the interpretation around selected locked cases. The revised material now gives the reader a more direct account of what was changed and how the evidence should be read.

The evidence now compares EnergyPlus- and Radiance-component outputs against analytic target values

for those cases. The stress test identifies weak agreement and ranking fragility; it does not support direct physical transfer of optimizer rankings.

The response therefore treats the terminology change as a claim-boundary correction. It explains why the calculation is still useful, but only as a cross-model stress test of selected cases rather than as broad validation.

Revisions made in the manuscript.

We made three corresponding revisions. First, we replaced validation-style physical-probe wording with limited physics-based cross-model stress test. Second, we revised Table 2 and Section 2.6 to state the source, output, and boundary of the physical cases. Third, we rewrote the Results and Discussion around weak agreement rather than confirmation.

Relevant revised manuscript and supporting evidence.

Main manuscript, Table 2 excerpt.

Mode	Source	Output	Boundary
Analytic response generation	generated descriptors	EUI _t , EG, H	analytic surface
Surrogate prediction	12 descriptors	DNN outputs	selected checkpoint
Feasible projection	descriptor candidate	nearest block	nearest neighbour
Physical stress test	locked blocks	physical outputs	24 cases
Climate sensitivity	four blocks	weather deltas	no retraining

Revised Table 2 excerpt. Evaluation modes are separated by source, output, and interpretation boundary.

Main manuscript, Section 2.6 paragraph.

The limited physics-based cross-model stress test uses 24 locked cases. Eighteen direct feasible cases compare analytic response-generator values with physics-based outputs without optimizer projection, while six optimizer-linked cases decompose the path from descriptor-space candidate to nearest generated feasible block and then to physics-based result. The direct and optimizer-linked subsets answer different questions and are not merged into a single parity plot.

Main manuscript, Section 3.5 paragraph.

The physics-based cross-model stress test completed all locked cases and provides a case-level comparison between analytic target values and physics-based outputs. Figure 9 uses only the 18 direct feasible blocks, so the parity plots are not affected by optimizer projection. The three targets show limited metric and rank consistency in this case set.

Supplementary Material, Section S5.4 paragraphs.

The locked physical workflow did not produce annual roof-plane irradiance outputs for the direct stress-test cases. The unavailable roof-irradiance pathway is therefore kept separate from the completed EUI_t, EG proxy, and H calculations.

EG is summarized with the GHI-based rooftop-PV proxy in the stress-test table. This proxy preserves a generation-related comparison across the locked cases, but it should be interpreted as a simplified rooftop-PV indicator rather than a full annual roof-irradiance result.

■ **Comment R2-3.**

The validation evidence remains insufficient. Only a small-batch probe is used, and the physical results show noticeable deviations from surrogate predictions, especially for EUIt and H.

Response.

We thank the reviewer for this comment. We agree that the previous validation evidence was insufficient and too optimistic. This point matters because the physical calculations should be reported as weak agreement where the tested cases show that result.

We revised Sections 2.6, 3.5, and 4.2, Fig. 9, Table 6, and Supplementary Section S5 to report completed physical cases, direct feasible-case metrics, and the downgraded interpretation. The revised material now gives the reader a more direct account of what was changed and how the evidence should be read.

The evidence now shows 18 direct feasible cases with EUIt MAE 3.788, nMAE 0.340, and Spearman rho -0.404; EG proxy MAE 0.441, nMAE 0.405, and rho -0.023; and H MAE 1.108, nMAE 0.601, and rho -0.043. These values support a stress-test conclusion about fragility, not physical equivalence or optimizer-ranking transfer.

This is why the response gives the numerical agreement values directly. The revision does not hide the weak correlations; it uses them to narrow the manuscript's conclusion and to prevent ranking-transfer language from reappearing.

Revisions made in the manuscript.

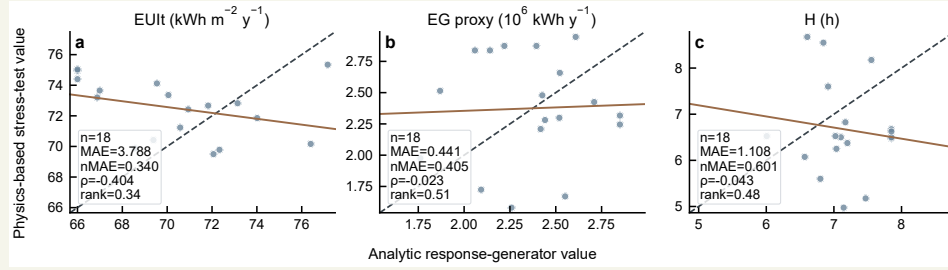
We made four corresponding revisions. First, we added the completed-case count and direct feasible-case summary. Second, we reported MAE, nMAE, and Spearman rho values in Table 6 and Fig. 9. Third, we revised Supplementary Section S5 to give case-level evidence. Fourth, we rewrote the Discussion to state that physical agreement is weak.

Relevant revised manuscript and supporting evidence.

Main manuscript, Section 3.5 paragraph.

The physics-based cross-model stress test completed all locked cases and provides a case-level comparison between analytic target values and physics-based outputs. Figure 9 uses only the 18 direct feasible blocks, so the parity plots are not affected by optimizer projection. The three targets show limited metric and rank consistency in this case set, with the weakest ordering stability appearing in the sunlight-related and demand-related comparisons. This result does not diminish the value of the stress test; instead, it clarifies the degree to which the analytic response surface and the physics-based calculations differ for selected generated blocks.

Main manuscript, Fig. 9.



Revised Fig. 9. Cross-model stress test for 18 direct generated feasible blocks.

Main manuscript and Supplementary Material, physical summary excerpt.

Target	Count	MAE	nMAE	Spearman ρ
EUI	18	3.788	0.340	-0.404
EG GHI proxy	18	0.441	0.405	-0.023
H	18	1.108	0.601	-0.043

Revised Table 6 and Table S4 excerpt. The direct feasible stress-test cases show limited metric and rank consistency.

Supplementary Material, Section S5.2 paragraph.

The physics-based stress-test set contains 24 locked cases. The 18 direct feasible cases are used to compare analytic response-generator values with physics-based outputs without optimizer projection. The six optimizer-linked cases decompose the path from descriptor-space candidate to nearest generated block and then to physics-based result. These two subsets should not be merged into a single parity plot because they answer different questions.

Comment R2-4.

The manuscript claims that the final benchmark uses the most accurate 2000-sample surrogate selected from a 500/1000/1500/2000-sample scale study. However, the sample-generation strategy is not sufficiently described. The authors should specify whether sampling was random, Latin hypercube, grid-based, constraint-based, or derived from feasible morphology generation. The distribution and coverage of the 12-dimensional design space should also be shown.

Response.

We thank the reviewer for this comment. We agree that the random feasible morphology generation procedure needed more detail. This point matters because the 2000 samples should be understood as generated feasible morphologies represented by derived descriptors, not arbitrary points in an unconstrained

descriptor cube.

We revised Section 2.1, Section 3.1, Table 1, Fig. 4, and Supplementary Sections S1.1–S1.5 to expand the generation protocol, descriptor coverage checks, 3 by 3 OSLI encoding, PCA summary, and duplicate checks. The revised material now gives the reader a more direct account of what was changed and how the evidence should be read.

The evidence now records prototype settings, descriptor distributions, six components for 95 percent descriptor variance, and absence of exact duplicate descriptor rows in the 2000-row canonical pool. This is why optimizer outputs are projected back to generated feasible blocks before morphology-level claims are made.

The response also connects generation to later projection. Because the optimizer searches descriptor space while the design object is a generated block, sample coverage and duplicate checks are necessary for interpreting feasible-morphology diversity.

Revisions made in the manuscript.

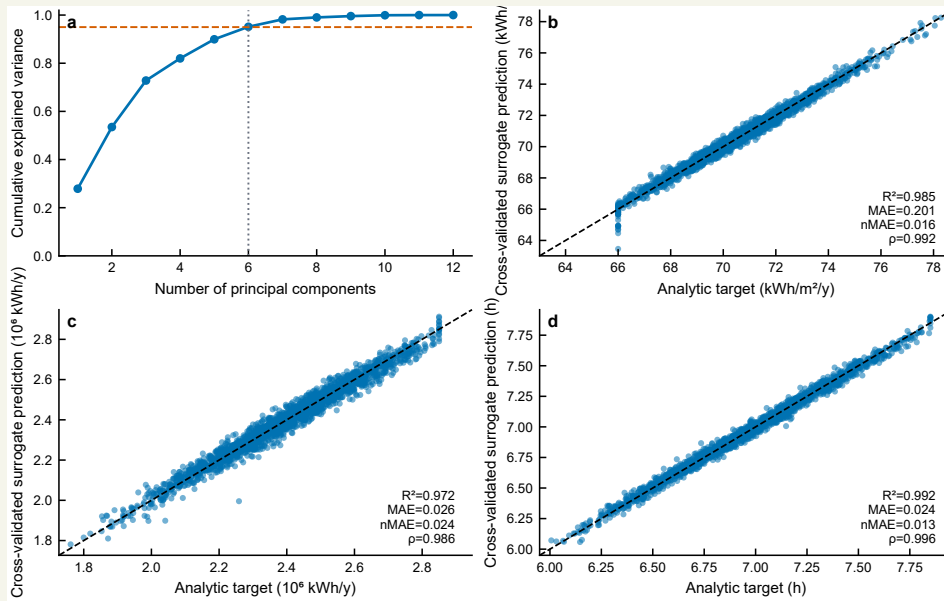
We made four corresponding revisions. First, we expanded the morphology-generation description in Section 2.1. Second, we revised Table 1 to define descriptors and OSLI encoding. Third, we added Fig. 4 and supporting distribution material. Fourth, we added Supplementary Section S1 material on prototype settings, PCA coverage, descriptor dependencies, and duplicate checks.

Relevant revised manuscript and supporting evidence.

Main manuscript, Section 2.1 paragraphs.

The morphology generator first constructs feasible block layouts and only then calculates descriptor values. Each block is represented as a $240\text{ m} \times 240\text{ m}$ square divided into a 3×3 grid of $80\text{ m} \times 80\text{ m}$ land units. One cell is assigned as open space, and the remaining eight cells are populated with residential prototypes. For each generated block, the generator samples the open-space position, building prototype, number of floors, and orientation angle. The orientation angle is sampled within -45° to 45° .

Main manuscript, Fig. 4.



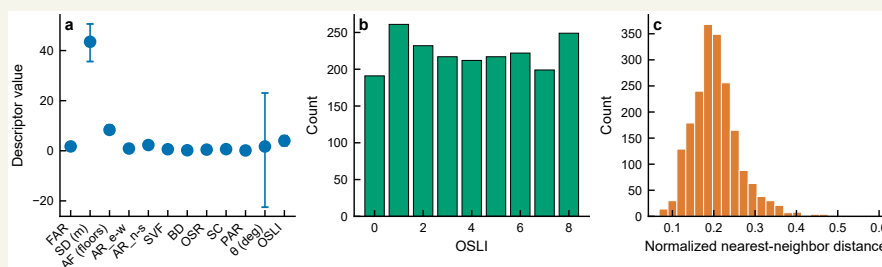
Revised Fig. 4. Descriptor coverage and cross-validated surrogate predictions.

Supplementary Material, Section S1.2 paragraph.

For each generated block, the open-space index is sampled from 0 to 8, theta is sampled uniformly from -45 deg to 45 deg, and each populated cell receives a randomly sampled prototype and a floor count sampled within that prototype range. Descriptor values are calculated only after this geometry-generation step. This order keeps the generated feasible block as the primary object and treats the 12 surrogate inputs as derived morphology descriptors.

Supplementary Material, Table S2 and Fig. S1 excerpts.

Check	Value	Interpretation
Exact duplicate count	0	no exact duplicate descriptor rows
Nearest-neighbour distance median	0.199	sparse normalized descriptor space
PCA components for 95% variance	6	structured descriptor space



Comment R2-5.

The RL formulation remains questionable because the problem is essentially a static black-box optimization problem rather than a naturally sequential decision-making process. The state transition is determined by the surrogate response to a morphology vector, and the sequential meaning of episodes and steps is unclear.

Response.

We thank the reviewer for this comment. We agree that the DDPG static black-box search formulation needed to be transparent. This point matters because generic RL language can mislead readers if the static surrogate-query setting is not stated.

We revised Sections 1.2, 2.4, and 3.2, Figs. 2, 3, and 6, and Supplementary Section S3.1 to define state, action, episode, reward, actor, critic, replay buffer, and target networks for the static surrogate benchmark. The revised material now gives the reader a more direct account of what was changed and how the evidence should be read.

The evidence now states that one episode is a sequence of 40 surrogate queries and that the reward is computed from the surrogate-predicted normalized target vector. DDPG is included as a policy-learning comparator under a matched query budget, not as a model of dynamic building operation.

This is why the response names each RL component instead of only citing DDPG. The component definitions show exactly where the formulation departs from physical time-series control and why the comparison remains a surrogate-query benchmark. It also explains why the episode count, query budget, and reward definition are reported together rather than treated as separate implementation details.

Revisions made in the manuscript.

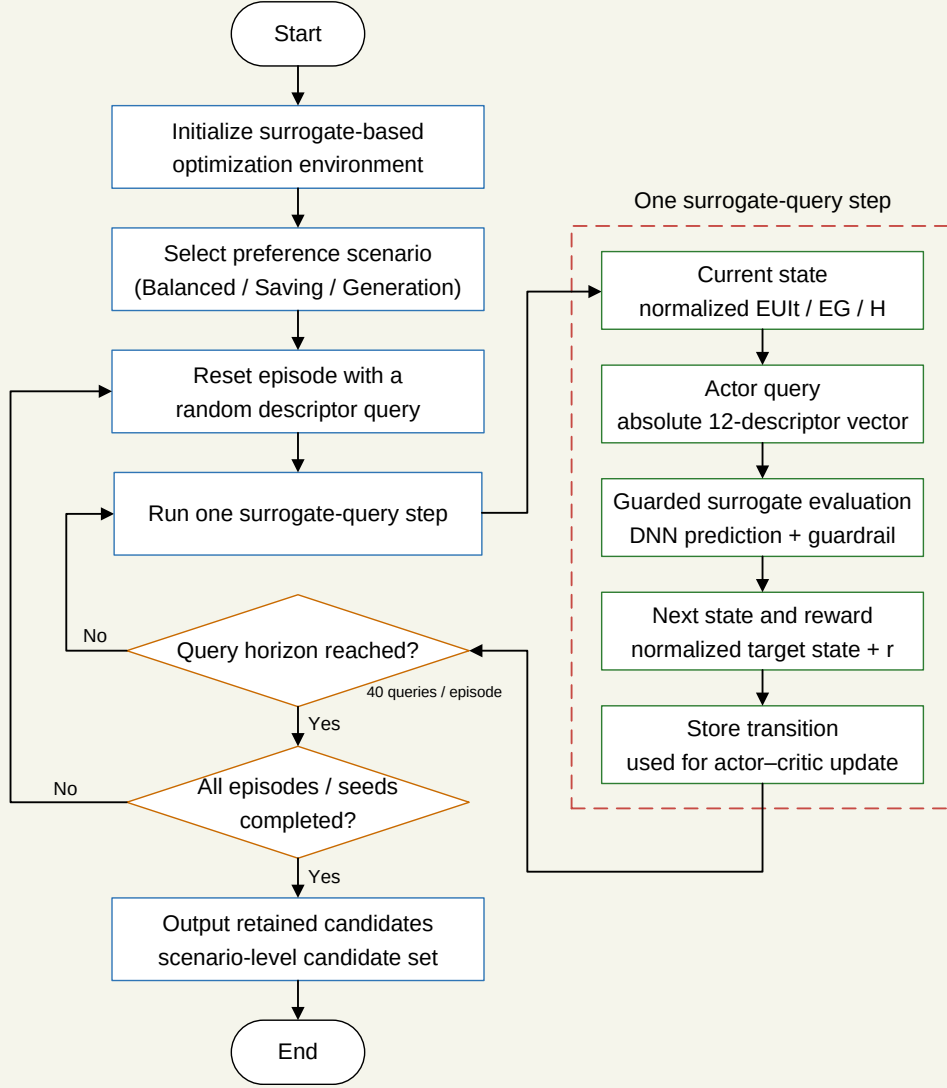
We made three corresponding revisions. First, we rewrote the DDPG method section around state, action, episode, reward, actor, critic, and target networks. Second, we revised Figs. 2, 3, and 6 to align with static surrogate-query interpretation. Third, we added Supplementary Section S3.1 for seed-level training behaviour.

Relevant revised manuscript and supporting evidence.

Main manuscript, Section 2.4 paragraph.

The DDPG environment serializes repeated queries to a static guarded surrogate. At the beginning of each episode, a random descriptor query is evaluated to construct the initial normalized three-target state. At each subsequent step, the actor outputs an absolute 12-dimensional normalized descriptor action. This action is not an incremental modification of the preceding block. The action is denormalized to descriptor bounds, evaluated by the guarded surrogate, and converted to the next normalized target state and scalar reward. The 40-step horizon therefore functions as a fixed sequence of policy-training queries on a static surrogate response surface.

Main manuscript, Fig. 2.



Revised Fig. 2. Serialized DDPG surrogate-query workflow. Each episode contains 40 absolute descriptor queries.

Main manuscript, reward equations.

$$z_i = \frac{y_i - y_i^{\min}}{y_i^{\max} - y_i^{\min}}.$$

$$\mathbf{u} = [0, 1, 1]^T.$$

$$d_w = \frac{\|\mathbf{w} \odot (\mathbf{z} - \mathbf{u})\|_2}{\|\mathbf{w}\|_2}, \quad R = 1 - d_w.$$

Supplementary Material, Section S3.1 paragraph.

Each DDPG scenario uses 20 seeds, 600 episodes, and 40 surrogate interactions per episode. The seed-level summaries record best reward, final reward, plateau episode, and best-to-final regression. These quantities describe policy-training behaviour under serialized surrogate queries on a static surrogate response surface.

■ Comment R2-6.

The manuscript uses several terms that need clearer distinction, including "analytic response-generation workflow," "independent analytic reevaluation," "physical probe," "full physical stack," and "publication-grade physical closure." These terms are currently confusing. A table comparing these evaluation modes would improve transparency.

Response.

We thank the reviewer for this comment. We agree that evaluation terminology was inconsistent. This point matters because readers need to know whether a result comes from analytic target assignment, surrogate prediction, projection, selected physical calculation, or climate sensitivity.

We revised Table 2 and Sections 2.2, 2.6, 3.5, and 4.2 to standardize the evaluation terms and assign each one a source, output, and interpretation boundary. The revised material now gives the reader a more direct account of what was changed and how the evidence should be read.

The evidence now uses analytic response generation for EUIt/EG/H target assignment, surrogate prediction for the DNN layer, feasible projection for descriptor-to-block mapping, physical stress test for locked cases, and climate sensitivity for weather-file changes. This terminology prevents overclaiming by keeping evidence layers separate.

This terminology cleanup also supports the reviewer-response evidence blocks because each displayed excerpt now has a named evidence source: analytic formula, surrogate result, projection diagnostic, physical stress test, or climate sensitivity.

Revisions made in the manuscript.

We made three corresponding revisions. First, we rewrote Table 2 as an evaluation-mode table. Second, we aligned Sections 2.2, 2.6, 3.5, and 4.2 with the same terminology. Third, we removed ambiguous wording that could conflate analytic response assignment with physical simulation or broad validation.

Relevant revised manuscript and supporting evidence.

Main manuscript, Table 2 excerpt.

Mode	Source	Output	Boundary
Analytic response generation	generated descriptors	EUIt, EG, H	analytic surface
Surrogate prediction	12 descriptors	DNN outputs	selected checkpoint
Feasible projection	descriptor candidate	nearest block	nearest neighbour
Physical stress test	locked blocks	physical outputs	24 cases
Climate sensitivity	four blocks	weather deltas	no retraining

Revised Table 2 excerpt. Evaluation modes are separated by source, output, and interpretation boundary.

Main manuscript, Section 2.6 paragraph.

Descriptor-space optimizer outputs are treated first as candidate descriptors. Each retained descriptor-space candidate is therefore mapped to the nearest generated feasible block in normalized descriptor

space. This mapping quantifies projection distance, duplicate collapse, and changes in HV and IGD under the fixed normalized reference front. The projection step is a representation check that precedes morphology-level interpretation.

Main manuscript, Discussion boundary paragraph.

These two sensitivity analyses do not convert the descriptor-space comparison into a physical ranking result. They instead define where the analytic response surface, generated feasible blocks, and selected physical workflows agree only partially, which is the boundary needed for early-stage design-support interpretation.

■ **Comment R2-7.**

The objective definitions require further clarification. EUI_t, EG, and H are said to conceptually correspond to EnergyPlus, rooftop PV, and sunlight-hour calculations, but the actual values are generated analytically. The authors should provide clear formulas or workflows for calculating each target in the generated dataset.

Response.

We thank the reviewer for this comment. We agree that the objective formulas and clipping ranges needed to be shown. This point matters because reproducibility and interpretation require the reader to see how EUI_t, EG, and H are generated for the analytic dataset.

We revised Section 2.2, Table 2, and Supplementary Section S1.4 to add the analytic response-generation equations, stochastic terms, objective directions, and clipping ranges. The revised material now gives the reader a more direct account of what was changed and how the evidence should be read.

The evidence now defines the orientation penalty, aspect-ratio average, openness and solar-access proxy terms, density penalty, response-generation noise, and final clipped outputs. The formulas define a controlled benchmark layer and are not claimed to be a full physical energy model.

The response also states the remaining boundary: exposing the formulas improves reproducibility of the analytic benchmark, but those equations are still target-assignment rules rather than a calibrated physical simulation model. This is why the response keeps both the equations and the clipping ranges in the evidence package: the formulas define the target surface, while the ranges define the benchmark values seen by the surrogate and optimizers.

Revisions made in the manuscript.

We made three corresponding revisions. First, we added the EUI_t, EG, and H analytic formulas to Supplementary Section S1.4. Second, we added the clipping ranges and stochastic-term convention. Third, we revised Section 2.2 and Table 2 so that the formulas are identified as analytic targets used for surrogate training and optimizer comparison.

Relevant revised manuscript and supporting evidence.

Main manuscript, Section 2.2 paragraph.

The response directions follow the design objectives: lower EUIt is preferred, while higher EG and H are preferred. The active 2000-sample pool is the largest nested sample regime used for the selected DNN surrogate. The full equations, station-bias convention, stochastic terms, and clipping ranges are given in Supplementary Section S1. Table 2 distinguishes analytic response assignment, surrogate prediction, feasible projection, and the post-optimization sensitivity analyses.

Supplementary Material, Section S1.4 equations.

$$\begin{aligned} p_\theta &= |\theta|/45, & a_m &= 0.5(AR_{ew} + AR_{ns}), \\ s &= \exp\left(-\frac{(FAR - 1.7)^2}{0.6}\right) (0.45 + SVF)(0.3 + OSR), & p_d &= \max(FAR - 2.8, 0). \\ EUIt^* &= 78.0 + 10.0BD + 4.3PAR + 2.8a_m + 0.022SD + 0.8SC + 2.1p_\theta \\ &\quad - 6.4OSR - 5.8SVF - 0.9AF - 4.5s + \epsilon_{EUIt}, \\ EG^* &= 0.96 + 0.28FAR + 0.64SVF + 0.52OSR - 0.19BD - 0.12a_m - 0.18p_\theta \\ &\quad + 0.26OSLI^* + 0.78s - 0.1p_d + \epsilon_{EG}, \\ H^* &= 6.45 + 1.1SVF + 0.55OSR + 0.48OSLI^* + 0.24s - 0.7p_\theta - 0.45PAR - 0.24a_m + \epsilon_H. \\ EUIt &= \text{clip}(EUIt^*, 66.0, 96.0), & EG &= \text{clip}(EG^*, 1.2, 2.85), & H &= \text{clip}(H^*, 6.0, 7.85). \end{aligned}$$

■ Comment R2-8.

The paper should provide code, data, or at least detailed reproducibility information.

Response.

We thank the reviewer for this comment. We agree that the reproducibility statement needed to be clearer about public and excluded material. This point matters because the response package should identify what can be released without exposing credentials, local logs, or restricted raw inputs.

We revised the Data and code availability statement, README, and response manifest to rewrite the availability wording around a sanitized public GitHub release. The revised material now gives the reader a more direct account of what was changed and how the evidence should be read.

The evidence now states that scripts and processed tables for the analytic response dataset, surrogate assessment, optimizer comparison, feasible projection, physical stress-test summaries, and cross-climate sensitivity summaries will be made available. Raw credentials, private logs, machine-specific paths, and non-redistributable inputs remain excluded.

This wording is intentionally practical. It tells the reviewer what can be audited from the public release

while avoiding any promise to publish credentials, private paths, or raw material that cannot be redistributed.

Revisions made in the manuscript.

We made three corresponding revisions. First, we rewrote the Data and code availability statement. Second, we aligned the response package README and manifest with the files used for this rebuttal. Third, we preserved the exclusion boundary for credentials, local logs, private paths, and restricted raw inputs.

Relevant revised manuscript and supporting evidence.

Main manuscript, Data and code availability.

The code, processed datasets, and result tables supporting the numerical analyses will be made available through the project repository at <https://github.com/haolinwang1313/DRL-Driven-Optimization>. The public release will include the scripts required to regenerate the analytic response dataset, surrogate assessment summaries, optimizer comparison tables, feasible-projection results, physics-based stress-test summaries, and cross-climate sensitivity results.

► Reviewer 3.

We thank Reviewer 3 for the comments on workflow-figure spelling, utility interpretation, reward formulation, unit consistency, and analytic/physical wording. These comments helped align the figures, equations, and claims with the revised evidence.

■ Comment R3-1.

Correct spelling and wording errors in the workflow figures. For example, Fig. 1 still contains visible typos such as "loactions," "Acotr network," and "Critic network," which should be revised to "locations," "Actor network," and "Critic network."

Response.

We thank the reviewer for this comment. We agree that the figure spelling and label errors needed correction. This point matters because workflow and architecture figures define the method structure, so visible errors reduce clarity.

We revised Fig. 1, Fig. 3, and their captions to correct the typo and align figure labels with the revised methodology and DDPG terminology. The revised material now gives the reader a more direct account of what was changed and how the evidence should be read.

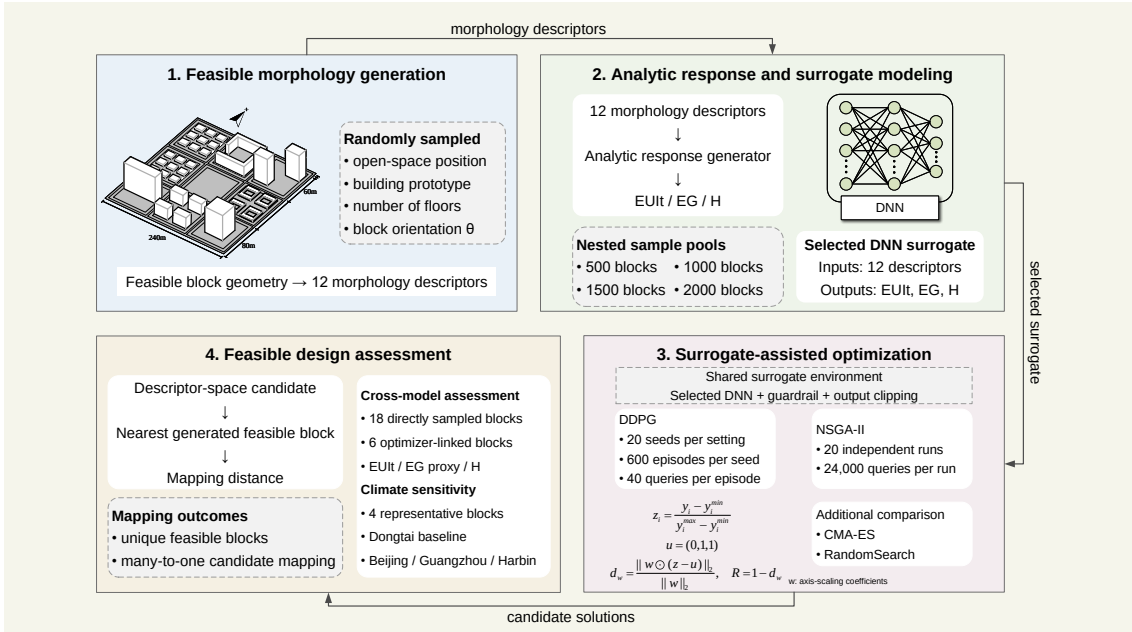
The evidence now Fig. 1 now uses consistent workflow-module labels, while Fig. 3 uses standardized labels for actor, critic, replay buffer, target networks, state, action, and reward path. This is a presentation correction and does not change the method or results.

Revisions made in the manuscript.

We made two corresponding revisions. First, we corrected the typo and terminology issues in Fig. 1 and its caption. Second, we corrected Fig. 3 and its caption so that the DDPG labels match the revised method description.

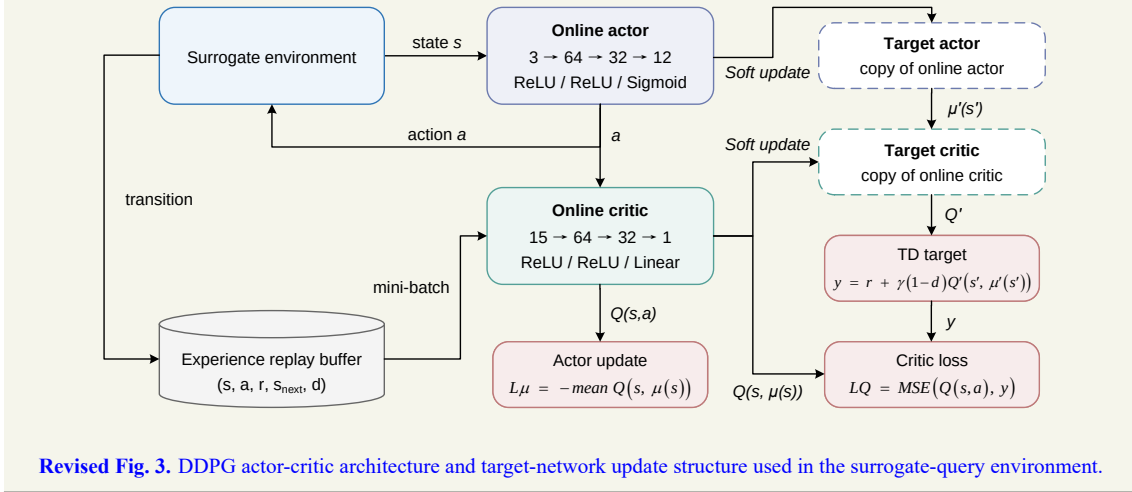
Relevant revised manuscript and supporting evidence.

Main manuscript, Fig. 1.



Revised Fig. 1. Workflow linking feasible morphology generation, analytic response assignment, surrogate-assisted optimization, feasible-block projection, and sensitivity analysis.

Main manuscript, Fig. 3.



Revised Fig. 3. DDPG actor-critic architecture and target-network update structure used in the surrogate-query environment.

Comment R3-2.

Verify the consistency between Fig. 9(d) and the stated benchmark conclusion. The text states that NSGA-II remains stronger than all DDPG scenarios, but the utility panel should be carefully checked to ensure that the plotted curves, legend, and utility direction support this conclusion.

Response.

We thank the reviewer for this comment. We agree that the earlier utility interpretation could be read as a direct optimizer-ranking result. This point matters because benchmark fairness requires separating

scalarized utility, equal-size HV/IGD, retained outputs, and feasible projection.

We revised Sections 3.3 and 3.4, Table 5, Figs. 7 and 8, and Supplementary Sections S3 and S5 to rewrite the comparison around fixed-reference descriptor-space metrics, equal-size diagnostics, and projection. The revised material now gives the reader a more direct account of what was changed and how the evidence should be read.

The evidence now shows NSGA-II strong in descriptor-space HV and IGD under the selected surrogate benchmark while DDPG remains preference dependent and projection changes feasible-block interpretation. Projection and physical stress-test evidence prevent descriptor-space ranking from being read as direct physical performance ranking.

The response therefore avoids ranking language based on one utility view. It explains that each metric has its own contract and that the projection stage changes how descriptor-space performance should be interpreted.

Revisions made in the manuscript.

We made three corresponding revisions. First, we rewrote Sections 3.3 and 3.4 to separate scalarized candidate selection, descriptor-space HV and IGD, equal-size diagnostics, and feasible projection. Second, we revised Table 5 and Figs. 7–8 around those criteria. Third, we removed the previous utility-only interpretation from the main claim.

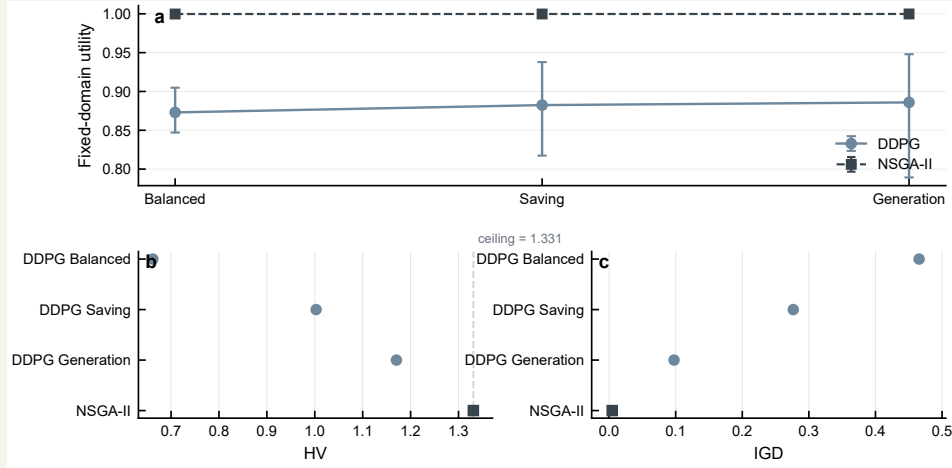
Relevant revised manuscript and supporting evidence.

Main manuscript, Section 3.3 paragraphs.

Table 5 summarizes descriptor-space performance under the fixed normalized reference front, while Fig. 7 separates fixed-domain utility from equal-size archive metrics. NSGA-II is close to the HV ceiling under the fixed reference and remains the strongest method in the equal-size HV and IGD comparison. Among the three DDPG settings, the Generation-focused scenario gives the strongest descriptor-space archive metrics, followed by the Energy Saving and Balanced settings.

The comparison should be interpreted through the retained-output structure. DDPG contributes one best scalarized candidate from each seed, whereas NSGA-II contributes final populations from repeated runs. The equal-size summary reduces, but does not eliminate, this structural difference. Therefore, the descriptor-space result supports a method-comparison statement under the selected surrogate and fixed reference; it does not by itself describe feasible-block diversity.

Main manuscript, Fig. 7.



Revised Fig. 7. Descriptor-space DDPG and NSGA-II comparison under fixed-domain utility and equal-size retained outputs.

Main manuscript, Table 5 excerpt.

Method	Scenario	HV	IGD	Note
DDPG	Balanced	0.661912	0.465522	scalarized output
DDPG	Saving	1.002757	0.276529	scalarized output
DDPG	Generation	1.170428	0.097551	scalarized output
NSGA-II	equal-size 20	1.330995	0.004710	common retained size
NSGA-II	full archive	1.330999	0.004947	2000 rows

Revised Table 5 excerpt. Descriptor-space metrics use a fixed normalized reference front.

Main manuscript, Section 3.4 paragraph.

The projection analysis shows that descriptor-space coverage and feasible-morphology diversity are distinct. NSGA-II retains 2000 descriptor candidates, yet nearest-neighbour projection maps them to 51 unique generated feasible blocks.

■ **Comment R3-3.**

Recheck the mathematical form of the weighted-distance reward in Eq. (5). The current expression appears to apply weights directly to the normalized objective values; if the intended formulation is a standard weighted Euclidean distance, the equation should be revised accordingly.

Response.

We thank the reviewer for this comment. We agree that the reward formula needed correction. This point matters because the previous expression could be misread as an artificial large-offset reward.

We revised Section 2.4 and Fig. 3 to define the reward directly from normalized weighted distance. The

revised material now gives the reader a more direct account of what was changed and how the evidence should be read.

The evidence now normalizes targets to z_i , uses the ideal vector $[0, 1, 1]^\top$, computes d_w with the preference vector, and sets $R = 1 - d_w$ with no large offset. This aligns the manuscript with the static surrogate-query implementation and makes the reward scale interpretable.

The response also explains why the correction matters: a bounded distance-based reward is interpretable across preference settings, while the removed offset could obscure the actual optimization signal.

Revisions made in the manuscript.

We made three corresponding revisions. First, we corrected the reward equations in Section 2.4. Second, we revised Fig. 3 so that the reward path follows normalization, ideal-vector comparison, weighted distance, and $R = 1 - d_w$. Third, we removed the confusing large-offset notation from the response and manuscript evidence.

Relevant revised manuscript and supporting evidence.

Main manuscript, Section 2.4 reward equations.

$$z_i = \frac{y_i - y_i^{\min}}{y_i^{\max} - y_i^{\min}}.$$

$$\mathbf{u} = [0, 1, 1]^\top.$$

$$d_w = \frac{\|\mathbf{w} \odot (\mathbf{z} - \mathbf{u})\|_2}{\|\mathbf{w}\|_2}, \quad R = 1 - d_w.$$

Main manuscript, preference-weight sentence.

The preference weights follow the target order [EUI_t, EG, H]: Balanced uses $\mathbf{w} = [1/3, 1/3, 1/3]^\top$, Energy Saving uses $\mathbf{w} = [0.6, 0.2, 0.2]^\top$, and Energy Generation uses $\mathbf{w} = [0.2, 0.6, 0.2]^\top$.

Main manuscript, reward boundary sentence.

This reward is separate from the fixed-domain post-hoc utility used later for benchmark comparison. The fixed-domain utility is used for post-hoc comparison, while R is the scalar signal used during serialized DDPG training.

■ Comment R3-4.

Further standardize units in figure axes and table headings. Terms such as EUI_t, EG, H, residuals, and morphology factors should consistently include units where applicable, for example "EUI_t (kWh/m²/y)," "EG (10⁶ kWh/y)," and "H (h)."

Response.

We thank the reviewer for this comment. We agree that target and morphology units were not consistently presented. This point matters because unit inconsistency makes it harder to audit results and compare figures.

We revised Tables 1, 3, 5, and 6, Figs. 4–10, and Supplementary tables and captions to standardize target units, morphology units, figure axes, and table headings. The revised material now gives the reader a more direct account of what was changed and how the evidence should be read.

The evidence now reports EUI as $\text{kWh m}^{-2} \text{y}^{-1}$, EG using the stated 10^6 kWh y^{-1} scale where applicable, H in hours, and morphology quantities by unit or type. The numerical results are unchanged; the revision improves auditability.

Revisions made in the manuscript.

We made two corresponding revisions. First, we standardized target and morphology units in the main figures and tables. Second, we aligned the Supplementary Information table and caption labels with the same unit convention.

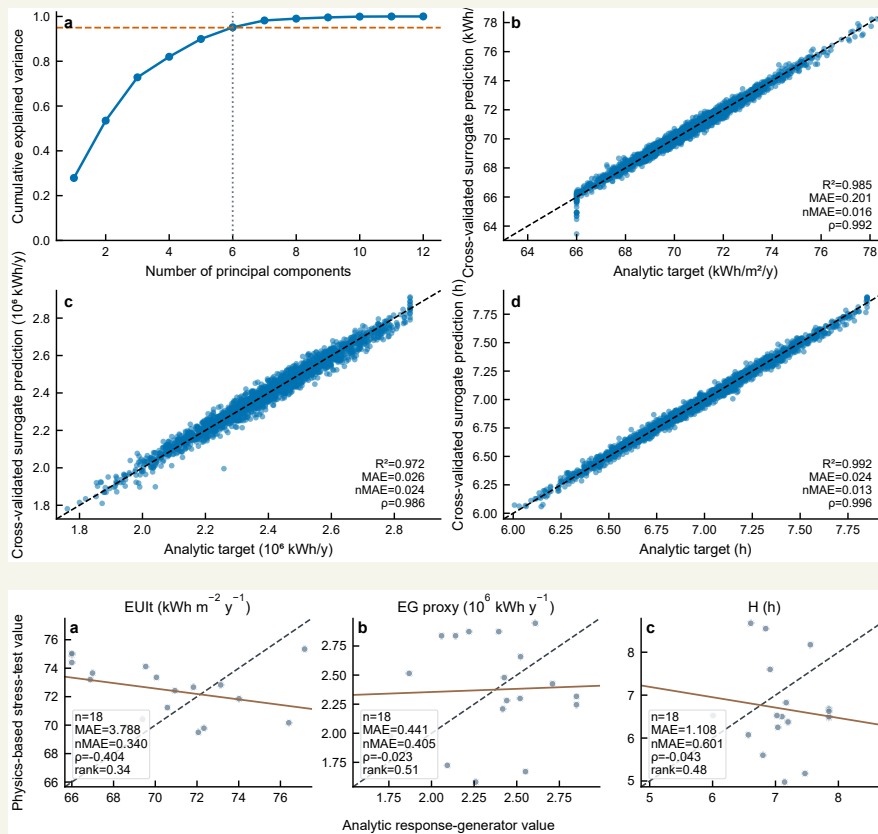
Relevant revised manuscript and supporting evidence.

Main manuscript, Table 1 descriptor units excerpt.

Descriptor	Unit or type	Construction	Interpretation
FAR	–	$A_{\text{gross}}/A_{\text{block}}$	density
SD	m	$h_{\text{max}} - \bar{h}$	height variation
AF	floors	$A_{\text{gross}}/A_{\text{foot}}$	linked to FAR, BD
AR_{ew}	–	$\bar{h}/W_{ew}$	east-west canyon
AR_{ns}	–	$\bar{h}/W_{ns}$	north-south canyon
SVF	–	analytic openness proxy	sky-view openness
BD	–	$A_{\text{foot}}/A_{\text{block}}$	footprint density
OSR	–	$A_{\text{open}}/A_{\text{gross}}$	open-space ratio
SC	–	$A_{\text{env}}/A_{\text{gross}}$	envelope exposure
PAR	m^{-1}	$P_{\text{sum}}/A_{\text{foot}}$	compactness
θ	deg	sampled in $[-45, 45]$	orientation
OSLI	integer	0–8	open-space cell

Revised Table 1 excerpt. The 12 quantities are derived morphology descriptors rather than independently sampled design variables.

Main manuscript, Fig. 4 and Fig. 9 unit-bearing evidence.



Revised figures. Target units are attached to axes and table headings where applicable.

Supplementary Material, Table S5 note.

ΔEUIIt is in $\text{kWh m}^{-2} \text{y}^{-1}$, ΔEG is in 10^6kWh y^{-1} , and ΔH is in h. Positive values denote increases relative to the baseline weather file.

Comment R3-5.

Make the wording about simulation, analytic approximation, and physical probing fully consistent. Since the manuscript states that the dataset is based on an analytic approximation rather than direct full-stack EnergyPlus/Radiance simulations, all related phrases should avoid implying that the final benchmark was performed using direct physical simulation.

Response.

We thank the reviewer for this comment. We agree that analytic, simulation, and physical wording needed consistent use. This point matters because the evidence source changes the interpretation of every optimizer and validation statement.

We revised the Abstract, Sections 1–4, Table 2, Table 6, and Supplementary Sections S1 and S5 to standardize analytic response generation, surrogate prediction, physical stress test, and climate sensitivity terminology. The revised material now gives the reader a more direct account of what was changed and how

the evidence should be read.

The evidence now makes clear that analytic response generation assigns targets to the generated pool, the surrogate predicts those targets during optimization, and selected locked blocks are examined through physical calculations and weather-file changes. This distinction supports the revised boundary: surrogate rankings are meaningful within the selected benchmark, while physical stress-test results show weak agreement and unsupported ranking transfer.

Revisions made in the manuscript.

We made three corresponding revisions. First, we standardized analytic response generation, surrogate prediction, physical stress test, and climate sensitivity terminology. Second, we revised Table 2 and Table 6 to carry the same evidence-source distinction. Third, we removed wording that could imply that the 2000 analytic samples were direct physical simulations.

Relevant revised manuscript and supporting evidence.

<u>Main manuscript, Abstract excerpt.</u>																											
Feasible block layouts are generated first and converted into 12 morphology descriptors. A morphology-to-performance response surface then assigns three targets to each sample: Energy Use Intensity, Energy Generation, and winter sunlight hours.																											
<u>Main manuscript, Table 2 excerpt.</u>																											
<table><tr><th>Mode</th><th>Source</th><th>Output</th><th>Boundary</th></tr><tr><td>Analytic response generation</td><td>generated descriptors</td><td>EUI_t, EG, H</td><td>analytic surface</td></tr><tr><td>Surrogate prediction</td><td>12 descriptors</td><td>DNN outputs</td><td>selected checkpoint</td></tr><tr><td>Feasible projection</td><td>descriptor candidate</td><td>nearest block</td><td>nearest neighbour</td></tr><tr><td>Physical stress test</td><td>locked blocks</td><td>physical outputs</td><td>24 cases</td></tr><tr><td>Climate sensitivity</td><td>four blocks</td><td>weather deltas</td><td>no retraining</td></tr></table>				Mode	Source	Output	Boundary	Analytic response generation	generated descriptors	EUI _t , EG, H	analytic surface	Surrogate prediction	12 descriptors	DNN outputs	selected checkpoint	Feasible projection	descriptor candidate	nearest block	nearest neighbour	Physical stress test	locked blocks	physical outputs	24 cases	Climate sensitivity	four blocks	weather deltas	no retraining
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<u>Supplementary Material, Section S5.4 paragraph.</u>																											
EG is summarized with the GHI-based rooftop-PV proxy in the stress-test table. This proxy preserves a generation-related comparison across the locked cases, but it should be interpreted as a simplified rooftop-PV indicator rather than a full annual roof-irradiance result.																											

► Reviewer 4.

We thank Reviewer 4 for comments on study necessity, climate-zone effects, energy-system expansion, best-scenario definition, and comparison with existing urban energy planning approaches. The revised manuscript strengthens the motivation while preserving the current model scope.

■ Comment R4-1.

There are plenty of studies related to block-scale urban energy planning, the necessity of this study needs to be further highlighted.

Response.

We thank the reviewer for this comment. We agree that the study necessity needed a clearer explanation. This point matters because block-scale planning decisions require early screening of urban form, energy demand, rooftop PV potential, and winter solar access before detailed system design is available.

We revised Sections 1.1–1.3 and the Discussion to rewrite the motivation, contribution, and positioning around representation-aware design support. The revised material now gives the reader a more direct account of what was changed and how the evidence should be read.

The evidence now shows why optimizer rankings depend on descriptor coverage, retained-output contracts, feasible-block projection, surrogate checkpoint, and evaluation mode. The contribution is a design-support and benchmarking contribution, not a universal superiority claim for DDPG or NSGA-II.

The response now ties necessity to the failure mode that motivated the revision: optimizer rankings can look precise while still depending on representation, surrogate reliability, and the projection from descriptors back to feasible blocks. This is the reason the response emphasizes benchmark fragility as a research need rather than treating optimizer comparison as a routine leaderboard exercise.

Revisions made in the manuscript.

We made three corresponding revisions. First, we rewrote the motivation and contribution paragraphs in the Introduction. Second, we added literature positioning for block-scale morphology, surrogate-assisted optimization, and policy-learning comparison. Third, we revised the Discussion so that study necessity is tied to representation-aware design support.

Relevant revised manuscript and supporting evidence.

Main manuscript, Motivation paragraphs.

Block morphology influences energy performance through coupled physical mechanisms. Density and height alter heating and cooling demand by changing envelope exposure, mutual shading, and micro-climatic context. Solar access and daylight availability depend on roof exposure, sky-view conditions, block orientation, and open-space arrangement. Recent large-scale studies further show that urban spatial

structure can shape building energy demand across broad urban areas and under extreme climate events, which makes morphology-sensitive design relevant beyond isolated building cases.

The computational challenge is that morphology-sensitive energy analysis becomes expensive when thousands of candidate blocks, multiple optimizer seeds, and several preference settings must be evaluated. Simulation-based urban form generation and building-performance optimization provide a strong basis for this task, but repeated full physical simulation remains costly in search loops.

Main manuscript, gap paragraph.

The remaining gap is methodological and representation-oriented. Urban energy optimization has established the value of morphology-aware design, surrogate modelling has reduced the cost of repeated evaluation, and reinforcement learning has proved useful in several energy-operation settings. What remains less clear is how policy learning and Pareto search should be compared when both operate on a shared surrogate response surface, when the surrogate inputs are descriptors derived from generated blocks, and when continuous outputs require projection before they can be interpreted as feasible morphologies.

Local references.

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■ **Comment R4-2.**

The selected location for modelling is in a specific climate zone. How this specific climate condition may affect the modelling and how the findings could be translated into other climate zones? This is not explained clearly.

Response.

We thank the reviewer for this comment. We agree that the climate-zone limitation needed direct treatment. This point matters because selected climate sensitivity is useful only if the manuscript states what it covers and what it does not cover.

We revised Sections 2.6, 3.5, and 4.2, Fig. 10, Table 6, and Supplementary Section S6 to add a four-block by three-climate sensitivity calculation and report weather metadata. The revised material now gives the reader a more direct account of what was changed and how the evidence should be read.

The evidence now shows stable EG ordering in the tested four-block set but climate-dependent EUIt and H responses, with the severe-cold case producing the largest EUIt increase. The surrogate is not retrained

by climate, and the analysis does not cover all climate zones or building types.

The response also states why the added analysis remains limited. It gives a climate sensitivity signal for selected locked blocks, but it does not claim that the surrogate or optimizer ranking generalizes across climate zones. The added wording also prevents climate sensitivity from being read as a new universal climate-transfer claim.

Revisions made in the manuscript.

We made three corresponding revisions. First, we added the four-block by three-climate sensitivity design. Second, we added Fig. 10, Table 6, Supplementary Table S5, and weather metadata. Third, we revised the Discussion to state that broader climate-zone coverage and climate-aware surrogate training remain future work.

Relevant revised manuscript and supporting evidence.

Main manuscript, Section 2.6 paragraph.

The baseline weather file used in the physical workflow serves as the reference condition for the cross-climate sensitivity analysis. The cross-climate sensitivity set then evaluates the same four block morphologies under three additional weather files representing Harbin, Beijing, and Guangzhou. The surrogate is not retrained, and the morphology set is not reselected after observing climate results.

Main manuscript, Fig. 10.

Revised Fig. 10. Cross-climate sensitivity of four selected blocks relative to baseline weather.

Supplementary Material, Section S6.1 paragraph.

The baseline weather file corresponds to the station used for the case-study physical workflow. The cross-climate sensitivity analysis evaluates the same four generated feasible blocks under Harbin, Beijing, and Guangzhou weather. The climate buckets are severe cold for Harbin, cold for Beijing, and hot-

summer warm-winter for Guangzhou. The TMY period is 2009–2023 for these weather files.

Supplementary Material, Table S5 excerpt.

Station	ΔEUI_t	ΔEG	ΔH	Main interpretation
Beijing	13.467	0.135	-0.056	higher demand
Guangzhou	-2.972	-0.057	-0.025	lower demand
Harbin	55.681	-0.044	-0.675	strong demand increase

Revised Table S5 excerpt. ΔEUI_t is in $\text{kWh m}^{-2} \text{y}^{-1}$, ΔEG is in 10^6kWh y^{-1} , and ΔH is in h.

Supplementary Material, Section S6.3 paragraph.

EG has stable ordering across the three additional climates, with Spearman ρ equal to 1.0 in each case. H shows lower ordering stability, with Spearman ρ values of 0.2 for Beijing, 0.4 for Guangzhou, and 0.2 for Harbin. EUI_t ordering remains stable for Beijing and Guangzhou but changes under Harbin, where Spearman ρ is 0.2.

■ **Comment R4-3.**

How the different sources of energy potentially affect the urban energy planning? This is specifically the case while China has rapidly developed the EV market. Can the distributed energy systems affect the urban energy planning?

Response.

We thank the reviewer for this comment. We agree that EV charging, storage, and distributed energy are important but outside the current objective system. This point matters because adding them as if they were modeled would create an unsupported claim.

We revised Section 4.3 and the objective-scope discussion to state the current objective scope and add these systems as future extensions. The revised material now gives the reader a more direct account of what was changed and how the evidence should be read.

The evidence now the current workflow covers feasible residential block morphologies, analytic EUI_t/EG/H target assignment, surrogate comparison, projection, and selected physical and climate-sensitive cases. Storage dispatch, EV charging demand, district-level grid exchange, tariff response, and distributed-energy operation are not modeled in the current revision.

This handling keeps the revision honest: the manuscript acknowledges these energy-system interactions as important, but it does not retrofit them into objectives or evidence that were not part of the current study.

Revisions made in the manuscript.

We made two corresponding revisions. First, we revised Section 4.3 to state that storage, EV charging,

distributed-energy operation, and district grid exchange are outside the current model. Second, we added these systems to future work as extensions that require new objectives and evidence.

Relevant revised manuscript and supporting evidence.

Main manuscript, Section 4.3 paragraph.

Several limitations remain. The optimizer comparison is conducted on an analytic response-generation dataset, while direct physical calculations are available for the locked stress-test cases. The 12 surrogate inputs are morphology-derived descriptors rather than independent unconstrained design variables. The DDPG horizon is best understood as a policy-training query sequence on a static surrogate. The cross-model stress test indicates limited rank consistency in the tested case set, and the climate analysis covers four blocks and three additional climates. Future work should connect optimization to a feasible morphology decoder, expand the physical simulation set, incorporate storage, EV charging, and distributed-energy interactions, and train climate-aware surrogates across multiple regions.

Main manuscript, objective scope excerpt.

The workflow generates feasible residential block morphologies, converts them into 12 morphology descriptors, assigns EUI, EG, and H through an explicit morphology-to-performance response surface, and trains a selected DNN surrogate from the 2000-sample regime of a nested scale study.

■ **Comment R4-4.**

It is not very clear what is considered as "best scenario" for optimization process.

Response.

We thank the reviewer for this comment. We agree that best scenario was ambiguous. This point matters because different criteria answer different questions and should not be collapsed into one label.

We revised Sections 2.5, 3.3, and 3.4, Tables 4–5, and Figs. 7–8 to separate best scalarized DDPG candidate, retained-output contracts, fixed-reference descriptor-space HV and IGD, equal-size comparison, feasible-block projection, and selected physical-case interpretation. The revised material now gives the reader a more direct account of what was changed and how the evidence should be read.

The evidence now shows that NSGA-II is strong in descriptor-space HV and IGD under the fixed benchmark reference, while DDPG outcomes are preference dependent and projection changes morphology-level interpretation. No single statement from these criteria defines a universal best physical scenario.

The response also clarifies the practical consequence for readers: a candidate can be strong under one scalarized or archive metric and still require projection and selected physical-case checks before design interpretation.

Revisions made in the manuscript.

We made three corresponding revisions. First, we rewrote Sections 2.5, 3.3, and 3.4 to define retained-

output contracts and comparison criteria. Second, we revised Tables 4–5 and Figs. 7–8 to separate scalarized candidates, archive metrics, and projection. Third, we removed ambiguous best-scenario wording.

Relevant revised manuscript and supporting evidence.

Main manuscript, Table 4 excerpt.

Method	Runs	Queries/run	Retained output
DDPG	20 seeds	24,000	best scalarized point
NSGA-II	20 runs	24,000	final populations
CMA-ES	20 runs	24,000	supplementary archive
RandomSearch	20 runs	24,000	supplementary archive

Main manuscript, Table 5 excerpt.

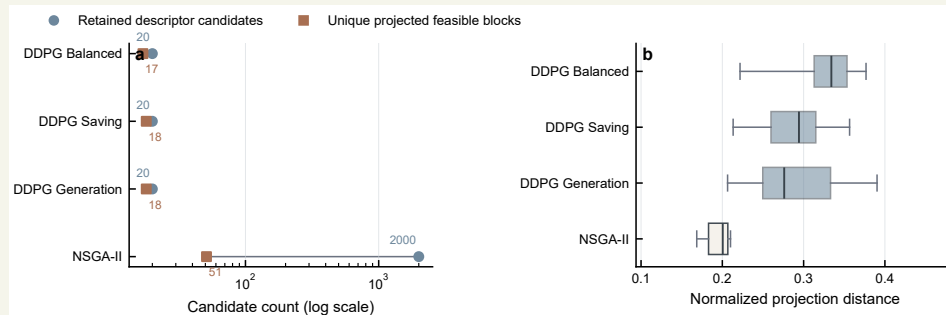
Method	Scenario	HV	IGD	Note
DDPG	Balanced	0.661912	0.465522	scalarized output
DDPG	Saving	1.002757	0.276529	scalarized output
DDPG	Generation	1.170428	0.097551	scalarized output
NSGA-II	equal-size 20	1.330995	0.004710	common retained size
NSGA-II	full archive	1.330999	0.004947	2000 rows

Revised Table 5 excerpt. Descriptor-space metrics use a fixed normalized reference front.

Main manuscript, Section 3.4 paragraph.

Projection distance adds a second layer of interpretation. The DDPG scenarios retain fewer descriptor candidates than NSGA-II, but their projected candidates are generally farther from the generated feasible blocks. In contrast, NSGA-II shows stronger descriptor-space coverage and shorter projection distances, yet its candidates still collapse to a much smaller feasible-block set.

Main manuscript, Fig. 8.



Revised Fig. 8. Descriptor-space candidate compression and projection distance after nearest-feasible-block mapping.

■ Comment R4-5.

More discussions are required to compare the proposed urban energy planning framework with existing approaches to validate its advantages.

Response.

We thank the reviewer for this comment. We agree that the manuscript needed a clearer comparison with existing approaches. This point matters because the comparison should explain the workflow's contribution without overstating its advantage.

We revised Sections 1.2, 4.1, and 4.3 to expand the comparison with simulation-based optimization, surrogate-assisted optimization, NSGA-II Pareto archives, DRL control studies, and neighbouring derivative-free methods. The revised material now gives the reader a more direct account of what was changed and how the evidence should be read.

The evidence now states the advantage narrowly as transparent separation of fast descriptor-space screening, feasible-block projection, and selected sensitivity analysis. The current evidence supports representation-aware early-stage design support, not comprehensive superiority over existing approaches.

The response therefore compares mechanisms rather than making a broad advantage claim. The stated contribution is that the workflow exposes representation and evaluation boundaries that can be hidden in conventional optimizer comparisons. This keeps the comparison useful for reviewers who want to know what the framework adds while preserving the limits imposed by the current evidence. It also names the remaining evidence needed for any stronger comparison.

Revisions made in the manuscript.

We made three corresponding revisions. First, we expanded the literature review to position the workflow relative to existing optimization and DRL studies. Second, we revised Section 4.1 to explain the representation-aware advantage. Third, we revised Section 4.3 to keep the comparison bounded by the current evidence.

Relevant revised manuscript and supporting evidence.

Main manuscript, literature review paragraph.

Parametric design studies show that geometry generation can support energy- and daylight-oriented design exploration. Research on parametric daylighting and low-energy design demonstrates that early-stage geometry and envelope choices can be formalized as searchable design spaces. Evolutionary multi-objective optimization remains widely used because it provides explicit trade-off archives for conflicting objectives. NSGA-II and related methods have been applied to envelope design, photovoltaic-oriented inverse urban design, urban block optimization, and energy-efficient shading design.

Main manuscript, DRL comparison paragraph.

In the energy domain, DRL has been widely adopted for HVAC control, building operation,

distribution-network control, and renewable-building operation. These applications are primarily operational control problems in which states evolve through time. Static urban morphology optimization differs from that setting because each query is an absolute candidate description evaluated by a surrogate response surface.

Main manuscript, Discussion paragraph.

For early-stage urban energy design, the workflow is useful because it separates fast descriptor-space screening from feasible-block projection and sensitivity analysis. A surrogate optimizer can identify promising descriptor regions, while projection and selected physics-based calculations help determine how those regions relate to generated block geometries. This separation makes the workflow more transparent for design-support use.

Local references.

- Nguyen, A.-T., Reiter, S., & Rigo, P. (2014). A review on simulation-based optimization methods applied to building performance analysis. *Applied Energy*, 113, 1043–1058. <https://doi.org/10.1016/j.apenergy.2013.08.061>
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- Ascione, F., De Masi, R. F., et al. (2016). Optimization of building envelope design for nZEBs in Mediterranean climate: Performance analysis of a residential case study. *Applied Energy*, 183, 938–957. <https://doi.org/10.1016/j.apenergy.2016.09.027>
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Closing

We again thank the editor and reviewers for the detailed comments. The revised manuscript and Supplementary Information now make the data-generation process, optimizer formulation, comparison protocol, physical and climate scope, and remaining limitations explicit. We hope that the revised package addresses the reviewers' concerns and provides a clearer basis for evaluating surrogate-conditioned optimizer comparisons in block-scale urban energy design.